

2020

UIC ENGINEERING



April 24, 2020

9:00 AM – 12:00 PM

Alternative Online Expo

engineeringexpo.uic.edu

ENGINEERING

Engineering has an extraordinary history of applying methods and technologies to overcome obstacles. Therefore, it was no surprise that our Senior Design faculty and advisors saw the task of bringing our annual UIC Engineering Expo online as an engineering problem, which they efficiently solved using the best technological tools available.

I am pleased to welcome you—virtually—to Engineering Expo 2020. This Expo takes place in the electronic spaces of shared drives, the teaching platform Blackboard, and the web rather than in person at the UIC Forum, but it packs all the student ingenuity and captivating ideas you have come to expect from us. I invite you to spend some time browsing engineeringexpo.uic.edu (choose the Expo 2020 tab) to see more than 200 student projects across 25 categories.

Special thanks go to the UIC Engineering alumni and friends who volunteered to serve as online judges for this year's engineering and computer science projects. Your feedback, guidance, and advice offer great value to our students as they move on from their Senior Design and Expo experiences into the work they will do after graduation.

I hope you enjoy this year's virtual showcase. If you see a project that impresses you, I encourage you to send the student team members some words of support on Twitter. Simply mention us @UICEngineering, and we will make sure to pass it on.



Pete Nelson, PhD
Dean of Engineering

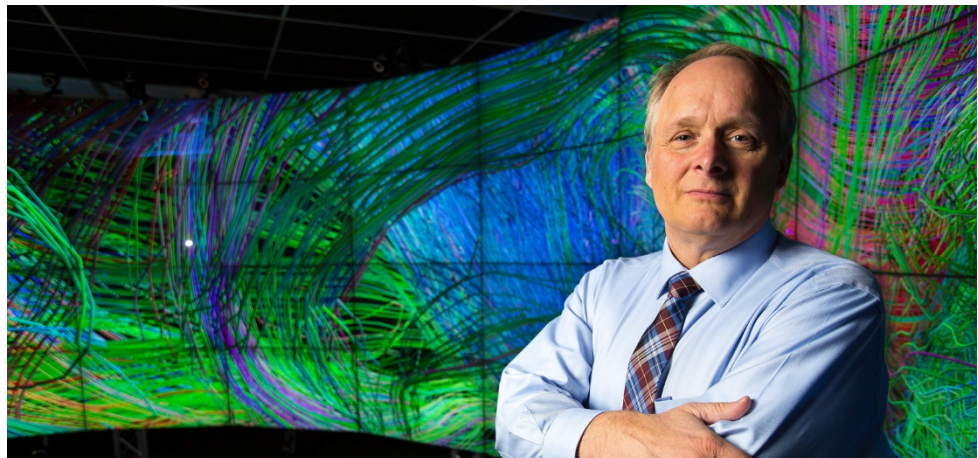


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Event Information

Schedule

Monday, April 20 through Friday, April 24	Online Presentations and Judging Score Submissions
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Friday, April 24 9:00AM – 12:00PM	Question and Answer Sessions with Judges and Students
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Origins

Engineering Expo began as the dream of an alumna, Kathy McGuire ('89). With the support of former Assistant Dean Arlene Norsym, her dream became a reality. Thirty years later, Expo has evolved into a dynamic capstone event. Senior Design students, with the guidance of faculty and industry advisors, develop a project from conception to product design, culminating with a presentation at Engineering Expo. This experience is crucial for our seniors as it merges group collaboration, engineering design, and commitment. With this event, our seniors take the first step in engineering their future and designing our world! We commend all participants for their hard work and flexibility in preparing their presentations for an online format and extend a special thank you to the judges for making an alternative Expo possible.

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BIOE.16 A Rapid Tear Recovery Device for Diagnostic and Investigative Purposes

Ameera Lodhi, Suha Mohiuddin, Naima Muckom, Farhin Patel, Yanyu Qian

Clinical Medical Devices

Adviser(s): Dr. Anthony E. Felder; Dr. Miiri Kotche

Sponsor: Dr. Sandeep Jain

Abstract: Dry-eye disease (DED) is a multifactorial disease which results in ocular dryness and discomfort. It results in proteomic, lipidomic, and metabolic changes to the ocular surface. Thus, analyzing the tear film provides diagnostic and prognostic assessment of DED and other related diseases. However, there is no standard method for collecting tears and current methods used by ophthalmologists are difficult, time-consuming, and resource demanding. For example, at UIC, only one highly-trained ophthalmologist analyzes tears, requiring a slit lamp, two research assistants, and eight glass capillary tubes. The design objective is to collect tear samples using a suction mechanism which directs collection of fluid directly into the Eppendorf tube. The proposed device has two components: a disposable handheld component applied to the eye during tear collection and a reusable tabletop component containing a 5V vacuum pump powered by two rechargeable AA batteries and switched on and off with a pushbutton. The handheld component is an airtight acrylic tube enclosure housing soft, medical-grade silicone tubing and an Eppendorf tube sealed with a complementary grommet and drilled plug. The vacuum pump creates negative pressure inside the handheld component, causing the sample to be suctioned directly into the Eppendorf tube. There is a duplicate pen component to avoid cross-contamination between eyes. Our device was required to collect at least 35 μ L of 50 μ L buffer solution, which is the minimum volume required for meaningful analysis. Testing was done on solutions with viscosities equivalent to those typical of tears. Preliminary testing demonstrated the device's ability to recover 80% of the sample volume. Collection time averaged 3-5 seconds depending on the viscosity of fluid. In conclusion, this device enables fast, high-yield collection of tear volume while minimizing patient discomfort. Further development of this class II device will allow for consistent, rapid tear sampling by a single minimally-trained practitioner.

BIOE.15 Automated Detection of Cancer in Prostatectomy Tissues by Elastic Modulus Measurements

Grieta Danosa, Veronica Gonzalez, Andrea Ohk, Vish Vijayakumar

Biomedical Engineering

Adviser(s): Dr. Anthony E. Felder; Dr. Miiri Kotche

Sponsor: Dr. Andre Kajdacsy-Balla; Dr. Virgilia Macias; Dr. Cristian Luciano

Abstract: Prostate cancer affects about 1.3 million men annually worldwide and is an area of active research. There is an unmet need for pathologists who receive prostatectomy samples from the operating room to accurately detect cancerous tissues within the sample while the cells are still viable for cell culturing and RNA sequencing. Currently, pathologists rely solely on manual palpation and visualization of the tissue to detect these tumors. Unfortunately, this approach has low sensitivity, detecting tumorous tissues less than 50% of the time. Previous research studies have designed palpation devices to diagnose cancer, but they do not assess tissue viability for xenografts nor localize the location of cancer with sufficient resolution to meet the need of pathologists. Here we propose a device to measure the elastic modulus and the relative differences in elasticity at different points of a prostate tissue slice with 1 cm resolution in 30 minutes to ensure the tissue remains fresh. The device is composed of a linear actuator with a force sensitive resistor, which is connected to an Arduino Uno. By displaying the relative differences in elasticity within the tissue sample, the pathologist would be better informed in determining the location of tumors and appropriate harvesting regions for research in patient-derived xenografts. We have verified the accuracy and reliability of our device on different pieces of rubber with known hardness with $p < 0.005$. Increased automation and accuracy in localizing the presence of the tumors will lead to better efficiency in obtaining viable tissue for research purposes to develop treatment plans for patients in the clinic.

BIOE.10 CNC-Adjustable Press Mold for Orbital Floor Reconstruction Surgery

Nelofer Arjumand, Heba Kanah, Shivani Senguttuvan, William Winski

Biomedical Engineering

Adviser(s): Dr. Miiri Kotche; Dr. Anthony E. Felder

Sponsor: Dr. Lee Alkureishi

Abstract: Orbital floor reconstruction surgery restores the form of the fractured floor and medial wall found in the maxillofacial skeleton. The surgery requires the insertion of a shaped titanium surgical mesh plate to provide structural support to the damaged bones of the orbital floor. A common cost-effective, yet time consuming method of shaping titanium plates is trial and error, where the surgeon attempts to identify the optimal anatomical position by repeatedly shaping a titanium plate. The proposed method attempts to shape titanium plates which can address the facial abnormalities of the eye for patients undergoing orbital floor reconstruction surgery to decrease the time and cost of patient specific models. An Anet A8 3D printer was modified such that the heating plate, cooling fans, and the safety protocol on the extruder motor were disabled to meet our design specifications. A machined pin bed containing 16 hexagonally packed cylindrical sockets was integrated with the modified 3D printer to hold pins that will generate a surface topology. Additionally, a script was written in G-code to extrude the X, Y and Z motor to each position of the pin bed. A second 3D printer will be used to house another pin bed, which will generate a mirror image of the first pin bed. The two pin beds press together to bend the titanium mesh plates required for the surgery. Next steps include using MATLAB to design an STL manipulation algorithm to output G code commands which will generate a contoured topology. The titanium plates will be shaped according to this patient-specific contoured topology, allowing it to be used in surgery. The solution enables physicians to decrease their time in the operating room without compromising accuracy when performing this surgical procedure.

BIOE.4 Cystoflush: Autonomous Intermittent Retrograde Pumping Irrigation System

Chloe Baratta, Dionna Bidny, Collin Dreilich, Jacob Gasienica

Clinical Medical Devices

Adviser(s): Dr. Miiri Kotche; Dr. Anthony E. Felder

Sponsor: Dr. Susan Talamini

Abstract: Continuous bladder irrigation (CBI) is used in the treatment of hematuria (the presence of blood in urine) to prevent the formation of blood clots within the bladder. CBI involves the cycling of irrigant solution, typically saline, into and out of the bladder to collect and remove blood accumulating in the organ. However, CBI is error-prone, with dangerous clot formation still occurring despite intervention. Clot formation can lead to irrigant retention and complications such as severe pain and bladder rupture, a life threatening emergency. A number of innovations have been developed to reduce clot formation and retention issues during CBI, but incorporate strictly reactive measures such as error alarm systems and inflow modulators to resolve clot hardening. These models are sub-optimal, as they only intervene following clot retention. Cystoflush augments traditional CBI with a retrograde pump designed to proactively agitate clot aggregates in the bladder, preventing the formation of large, hardened clots that cause obstruction. The prototype connects a programmable pump to the drainage channel of existing CBI. This system operates in two modes: "traditional CBI" mode and novel "pump" mode. Modes are defined by solenoid valves that control flow direction. A microcontroller software governs both valve orientation and bolus pumping frequency based on color intensity and opacity of outflow drainage fluid, measured by a colorimetric sensor. The capacity of the colorimetric sensor to differentiate between red color intensity and opacity for distinct blood dilutions was tested. The blood dilutions were formulated to represent different concentrations commonly seen in CBI outflow by practicing urologists. Testing revealed statistical differentiation between 87.5% of compared values. Correlation of the sensor output to pumping frequency is still reliant on additional testing and sensor improvement. With further innovation, the novel Cystoflush will greatly improve patient outcomes during CBI and lessen burden on healthcare providers globally.



BIOE.6 Design and Development of an Animal Positioning System for Preclinical MRI with 3-D Printing

Tarek Almardini, Osayd Nazzal, Muneeb Omar, Tarek Safieh

Biomedical Engineering

Adviser(s): Dr. Anthony E. Felder; Dr. Miiri Kotche

Sponsor: Dr. Weiguo Li

Abstract: Magnetic resonance imaging (MRI) scans on small animals such as mice have allowed for a greater biological understanding without subjecting humans to extensive medical testing and analysis. However, MRI imaging requires a stabilized body for an extended period of time, otherwise images are prone to motion artifacts. Thus, it is important to minimize movement of the subject as much as possible. Motion artifacts, due to either voluntary or involuntary movement, can appear as blurring, darkening of the scan, or may even mimic serious medical conditions in some instances. Current cradle solutions with a “one size fits all” design approach are not specific to one species and do not provide the desired image clarity. This can result in inadequate scans and positioning that is impossible to replicate through standard protocol and cannot be compared accurately across images. In addition, current options available on the market employ invasive methods such as tubes or bars placed in the mouse’s ears for head fixation. Therefore, we designed a mouse positioning system for preclinical MRI using 3D printing, a neonatal pressure cuff, and a customized bite bar to reduce involuntary motion during imaging. Performance criteria for our device included a twenty percent reduction in motion artifacts. Moreover, we demonstrate the testing procedure to perform such an experiment along with methods to analyze the results, a successful design will reduce motion artifacts by at least twenty percent.

BIOE.7 Early Warning System for Charlestown, Nevis

Kristen Kacich, Jared Miranda, J'Heannheh Belle Rosier

*Consumer Product Development -
Wellness Devices*

Adviser(s): Dr. Miiri Kotche; Dr. Anthony E. Felder

Sponsor: Dr. Janet Lin

Abstract: Caribbean residents are prone to a variety of disasters including hurricanes and floods. Early warning systems are used to convey risk knowledge, monitor disasters, give a response to disasters and communicate information in a succinct and actionable way. One of the most common and efficient early warning systems is a warning siren that gives audio messages to residents, however the Caribbean town of Charlestown, Nevis currently does not have a warning siren. An outdoor speaker was innovated to act as a warning siren with audio messaging capability. Current warning sirens on the market can only store a certain amount of pre-recorded audio messages and are very expensive. However, this warning siren was created using a low-cost speaker and amplifier, making this warning siren cheaper. The warning siren has a pre-recorded audio interface that includes hurricane and flood warnings, but also a live audio interface that allows the user to record their own messages. The interfaces were created using Arduino, Raspberry Pi and JavaScript. In order for the warning siren to be audible across Charlestown, the sound intensity of the output audio must be at least 98.84 dB from 100 ft away. In order to test that our prototype operates at the given frequency range, five trial runs were performed on both audio interfaces and the sound intensity of each trial run was collected. The standard deviation was calculated and we expect that 95% of the gathered data (two standard deviations away from the mean) to be at least 98.84 dB. Having a warning siren that has a great enough sound intensity will allow the residents of Charlestown to listen to warnings about any potential disasters and will lead to the residents being better prepared for when the disaster strikes.



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BIOE.9 FloSelect

Farhan Ali, Syed Ali, Deja Anderson-Fitzell, Gabriela Wojciak

*Consumer Product Development -
Wellness Devices*

Adviser(s): Dr. Miiri Kotche; Dr. Anthony E. Felder

Sponsor: Cheryl Evans; Vocatek

Abstract: The swallowing disorder Dysphagia can arise from several cases of medical issues ranging from strokes to degenerative neurological conditions; thus any person of any age can suffer from dysphagia. Current methods for treating dysphagia include conventional speech therapy, patient education and swallowing maneuvers such as tucking the chin; however, these methods only touch base on what patients should do to live safely with Dysphagia and in most cases they don't work. Our design FloSelect, is a consumer tabletop device that measures the flow rate of any liquid, while giving the user feedback in a yes/no reading, indicating if it is safe for them to drink the liquid or not in accordance with their speech pathologists prescribed safe swallowing range. The safe swallowing range is determined by a speech pathologist via the use of barium swallow studies. FloSelect is a 3D printed box that is comprised of a compartment at the top where the liquid is poured into for testing. To start the test the user presses a button on the LCD screen which signals the servo to lift a plexiglass wall that releases the liquid down a ramp at a steady rate for every test. FloSelect utilizes two Lux sensors under a four inch ramp that measure the time it takes for one liquid to cross the first to the second sensor. Once the testing is done, the LCD screen displays the final readings of either a yes or no reading to the user. FloSelect will be used as a means of a preventive method decreasing the chance for the liquid to go down the wrong path into the trachea which causes aspiration related injuries.

BIOE.11 Improved Cytology Brush Design for Diagnosing Oral Cancer

Akshara Acharya, Shreya Chaudhry, Monica Erramilli, Sandra Rivas

Clinical Medical Devices

Adviser(s): Dr. Miiri Kotche; Dr. Anthony E. Felder

Sponsor: Dr. Guy Adami; Dr. Joel Schwartz

Abstract: Oral cancer caused an estimated 9,750 deaths in the United States in 2010. Late diagnosis can increase the probability of oral cancer advancement which can be prevented by early analysis of visible precancerous lesions using oral cytology. Oral cytology is a non-invasive technique used to collect epithelial cells in the mouth to analyze for the presence of cancer genes. MASS aimed to design an oral cytology brush to collect a comparable yield of microRNA to existing brushes on the market while reducing lesion bleeding, which can contaminate the sample. Additionally, the brush design aims to satisfy the centrifuge tube size constraint and autoclavable sterilization requirement. After several design iterations, the final brush prototype has a head that is easy to remove for analysis, has no sharp edges, and is closer to fitting into a standard 1.5 mL centrifuge tube. The brush consists of an autoclavable handle and a brush head composed of soft nylon bristles (0.0762 mm diameter) adhered to a silicone base. To test the effectiveness of the prototype, the difference in the average microRNA yield between the prototype and control brush would be used to determine how similarly the two brushes perform using an equivalence test. CytoSoft™, the most common design in the market, would be used as the control group. Reducing the brush head diameter by simply trimming down the base and the silicone would allow the head to fit into a centrifuge tube for microRNA testing. The improved oral cytology brush design will provide an easier alternative to collect samples and aid clinicians in diagnosing oral cancer and diseases.

BIOE.3 Improving the Prenatal Care for Hearing Impaired Mothers

Allison Bayro, Daniela Merlos, Anna Metlushko, Claudia Stampa

*Consumer Product Development -
End User Medical Devices*

Adviser(s): Dr. Anthony E. Felder; Dr. Miiri Kotche

Sponsor: Dr. Aparna Ramanathan

Abstract: The current fetal Doppler, a staple instrument used in prenatal appointments, produces an auditory-only representation of fetal heartbeats. This is a limitation for Deaf individuals who cannot hear these tones and are excluded from the experience. This is an important consideration as a poor parental attachment in Deaf, pregnant women are closely related to adverse outcomes including preterm labor and low birth weight babies. Therefore, the aim of this project is to develop a multi-sensory fetal Doppler device for Deaf and otherwise challenged individuals. The product requirement designs (PRD) of this device consisted of a two-pound weight limit for portability, an outer coating of silicone for disinfecting purposes, a Bluetooth module for wireless transmission, and a vibrational motor and LED to emit tactile and visual outputs. The device consists of two components: a transmitter and a handheld heart visualizer. The first component attaches to the fetal doppler and uses an audio jack cord to capture audio output from existing Doppler devices. The embedded electronics contain a microcontroller, which analyzes the occurrence of heartbeats and relays this information to the handheld component via Bluetooth. This hand-held component consists of a heart-shaped design with embedded lights and vibrational motors to provide a multi-sensory representation of fetal heartbeats. The ability of the device to accurately detect and depict fetal heartbeats in real-time was tested successfully by comparing the calculated beats per minute (bpm) from the illumination of the hand-held device to the bpm displayed on the doppler screen. Future iterations of this device aim to incorporate wireless charging as well as a cell phone application to store, track, and monitor fetal heart tones throughout one's pregnancy. Thus, a Deaf pregnant mother is now able to experience the sensation of the baby's heartbeat.

BIOE.12 Multi-Electrode Contact Electroretinography (MECE)

Maram Dohal, Iram Hameeduddin, Khushal Shah, Natalie Vazquez

Biomedical Engineering

Adviser(s): Dr. Anthony E. Felder; Dr. Miiri Kotche

Sponsor: Dr. John Hetling; RetMap

Abstract: Electroretinography (ERG) is a diagnostic test performed to evaluate retinal function by measuring the electrical potential generated in the retina in response to a flash of light. The Contact Lens Electrode Array (CLEAr Lens), an existing ERG recording system created by John Hetling, analyzes spatial differences on the retina using multiple fluid channel electrodes on the cornea. While versions of CLEAr Lens exist for both humans and rats, with 32 and 25 electrodes respectively, this technology has not been developed for mice, despite their popularity as animal models used in research. Mice are preferred due to their lower cost, shorter time to reach maturity, and wider availability of genetic strains; however, with a corneal diameter of approximately 2.8 mm, a mouse cornea is too small to simply scale down the existing CLEAr Lens design. As such, a revised ERG recording lens, named the Multi-Electrode Contact Electroretinography (MECE) was developed to accommodate the size constraints of a mouse eye, for our project sponsor, Dr. Hetling, and our corporate sponsor, Ocuscience. The MECE Lens consists of a silicone-filled tube with an LED, a PCB to connect the electrodes to a data acquisition system, and five gold wires to contact the eye directly. Recording electrical signals is an important design requirement the MECE lens must fulfill, and in order to do so, the contact surface was molded to conform to the natural shape of the average mouse cornea using silicone, with gold wires placed in five specific corneal locations. Future testing will involve measuring impedance values across each pairing of electrodes, to ensure a proper fit, and recording electrical potential in response to the light stimulus to ultimately compare with expected ERG data and models.



Engineering Expo 2020

BIOE.8 Orthosis to Prevent Plagiocephaly in Infants

Rasean Hyligar, Ike Kabir, Jinan Mohammad, Ritu Shah

Consumer Product Development -

Adviser(s): Dr. Miiri Kotche; Dr. Anthony E. Felder

End User Medical Devices

Sponsor: Thaddeus Holgado

Abstract: Plagiocephaly, also known as “flat head syndrome,” is a condition that is prevalent in infants aged 0 to 4 months. The recent increase in plagiocephaly can be attributed to reducing the incidence of Sudden Infant Death Syndrome (SIDS) and an increased amount of time newborns spend in carrier devices. If untreated, plagiocephaly can lead to cognitive and motor delays, visual impairments, and noticeable facial asymmetry. Current devices attempt to conquer plagiocephaly by either correcting or preventing the deformation of the head by only reducing the pressure in a susceptible region, which does not fully address the problem. In contrast, our device addresses plagiocephaly by redistributing the pressure and decreasing the time a region of the head spends in one position, therefore preventing the onset of the condition altogether. The proposed solution allows for safe rotation of the infant's head, which can prevent the onset of plagiocephaly. Our device functions by utilizing an air pump that fills an array of inflatable compartments. Through an actuation mechanism, an Arduino microcontroller and servo motors work in tandem to activate valves, which allow the pumped air access to the compartments using preset time-based intervals. Air is transferred to each compartment through a semi-closed loop system. Air compartments in the mattress inflate in an alternating manner to rotate the infant's head 45 degrees off-axis from one side to another. The device complies with FDA class 2 standards for crib accessories. Our proposed device will benefit the growth of the infant and mitigate parent concerns.

BIOE.13 Single Use Syringe

Jakub Marek, Moin Owaisi, Riken Patel, Andrew Rivera

Clinical Medical Devices

Adviser(s): Dr. Anthony E. Felder; Dr. Miiri Kotche

Sponsor: Dr. Terry Layton

Abstract: In the current marketplace, the most commonly used syringe is a disposable model. The problem with the disposable syringe is that it can be misused to provide multiple injections due to negligence or to save cost. When a syringe is reused, the transmission risk of diseases increases. Some of these diseases include Hepatitis B, Hepatitis C, and HIV/AIDS. Currently, there are no types of single use syringes being sold in the mass market. However, some novel designs we found include the breakaway plunger, retractable needle syringe, single-use fillable syringe, and k1 syringe. These designs never made it to the mass market as they are too complex to make/use, and costly unit price. Our goal is to prevent the transmission of diseases/contamination by designing a syringe that purposely fails after a single injection. We accomplished this by implementing a mechanical design to remove the functionality of the plunger. Our design implements a ring embedded inside the barrel so that when the plunger is injected past this ring, any attempt to re-aspirate causes the rubber on the plunger to be removed which disables the syringe. Our Hi-Fi prototype consists of a ring made with 1 mm wire threaded through the barrel providing enough resistance to meet our design requirement. The most important aspect of a syringe to test are the forces applied for use. The glide force is the force required to push past the ring. The breakaway force is the pulling force required to break the rubber tip off the plunger. This is done to make our device user friendly and comparable to the widely used disposable syringe. Our final product will be created with injection molding including a negative CAD design of the barrel. The single use syringe will help prevent millions of cross contamination cases between patients.

BIOE.1 Skeletal Muscle Phantom for Elastography Imaging

Kazi Hussain, Aime Luna, Malek Qwaisni, Alex Rivera Hidalgo

Biomedical Engineering

Adviser(s): Dr. Miiiri Kotche; Dr. Anthony E. Felder

Sponsor: Dr. Thomas Royston

Abstract: Phantoms used in elastographic imaging to calibrate measurement systems and advance measurement methodology currently do not accurately characterize highly anisotropic tissues, such as striated skeletal and cardiac muscle. These tissue types also undergo significant changes in their mechanical properties due to the variations in mechanical loading they are subjected to as part of their normal function. Researchers need physiologically accurate, durable muscle phantoms to improve calibration systems for elastography, whether performed using optical, ultrasonic or magnetic resonance imaging modalities. Skeletal muscle phantoms should mimic the properties of real muscle's tissue in order to be used as a calibrator for measurement methodology in the detection, classification and monitoring of diseases. Our proposed solution is a multilayered phantom consisting of different materials that can be accurately used as a developmental tool in imaging processes. The modifications made to the original prototype bring upon a "pillar-in-pillar" design, an adaptation of Dr. Thomas Royston's previous work of pillared columns. Our design used material that we hypothesized would satisfy device requirements of anisotropy, density, and dimensions. Due to unforeseen circumstances, we were unable to proceed with testing that verifies mechanical properties of our novel design using Scanning Laser Doppler Vibrometry (SLDV) and Magnetic Resonance Elastography (MRE). Mock data was provided for the SLDV experiment, consisting of elastic modulus for our phantom test groups so we could conduct comparative studies on the different designs' viscoelastic properties. Our final design would be working toward a phantom that could be used across several modalities to develop and calibrate imaging methods. Possible next steps to improve future phantoms would be to incorporate a piezoelectric component to further characterize skeletal mechanical responses as well as verify anisotropic properties using MRE. Our work has the greatest impact on the development of future phantoms' properties and materials used in general elastography imaging.

BIOE.2 SPC Aide: Suprapubic Catheterization Placement Device

Abdel Bileomon, Kevine Boutcheko, Mateusz Drozd, Khyati Chauhan Singh

Clinical Medical Devices

Adviser(s): Dr. Miiiri Kotche; Dr. Anthony E. Felder

Sponsor: Dr. Michael J. Young

Abstract: Suprapubic catheterization (SPC) is bladder drainage completed by locating the bladder using an ultrasound transducer, and then inserting a catheter into the bladder through the lower abdomen. The procedure is considered "blind" because the transducer is removed after the bladder has been located, and the actual needle placement is performed without image guidance. If the needle misses the bladder it may puncture surrounding organs. We aim to provide accurate guidance of the needle into the bladder of patients undergoing this procedure to decrease the risk of injury or death. Our objective is to design a system incorporating bladder scanning technology supported by mechanical properties, and software to facilitate accurate guidance of the needle into the bladder. The SPC Aide stabilizes the transducer to provide uninterrupted visualization of the bladder throughout the procedure and supports the needle as it enters the abdomen. Our device determines the position of the transducer, calculates the optimal trajectory of the needle-insertion tube, and then moves it to that angle. A sensor determines the position of the transducer and the clinician enters the distance between the abdominal surface and the center of the bladder from the ultrasound screen into the SPC Aide. The required angle of the needle tube is calculated, and servos move the needle tube to the correct position. The physical attributes of the device include its adhesive wings, and a curved base for support and conformity to any abdominal surface. Slits in the needle-insertion tube enable the clinician to remove the device once SPC is completed. Our validation testing included verifying the resolution of the servos, and the accuracy of the device in targeting a point at a specific target distance with the transducer at different positions.

BIOE.5 Ultrasonic Radar System to Prevent Residential Drownings

Theo Hellmann, Xiang Li, Peiyao Li, Ryan Morley

*Consumer Product Development -
Wellness Devices*

Adviser(s): Dr. Anthony E. Felder; Dr. Miiri Kotche

Sponsor: Ryan Kole

Abstract: Drowning is a worldwide problem, and according to the CDC, 390 Americans drown in residential pools every year. Moreover, drowning is the number one cause of unintentional death for children between the ages of 1 and 4. To date, protective measures against drowning are education and anti-drowning devices. However, the existing devices on the market are limited, expensive, and often require swimmers to wear sensors on their body. Therefore, we have designed a cheaper drowning monitoring device that requires no worn sensors and can be used in residential pools. The device consists of an array of doppler radar sensors placed both above and below water to track swimmer location and activity. The device detects if a person is underwater too long which causes an alarm to go off and a notification to be sent to a cell phone. This device will enable peace of mind for swimmers and prevent drownings.

BIOE.14 Vari-Ox Bag

Michael Castillo, Humberto Dominguez Jr., Yash Nadkarni, Raj Patel

Clinical Medical Devices

Adviser(s): Dr. Anthony E. Felder; Dr. Miiri Kotche

Sponsor: Dr. Girish G. Deshpande

Abstract: The World Health Organization states that there were 6.9 million deaths relating to hypoxia and/or hyperoxia. Hypoxia is the condition in which the body is oxygen deprived, while hyperoxia is a condition in which the body has too much oxygen. In a direct countermeasure to these conditions, a new design for an oxygen blender has been proposed. These blenders are able to titrate oxygen from 100% to 21% in increments of 10%. These older devices, however, are bulk, reducing its portability. This causes complications when a patient has to be moved into a different location within the hospital. It is especially relevant when a patient needs to be resuscitated and transferred to an operating room. Many third world countries cannot afford nor do they have the electricity to power them. As a result, physicians tend to give 100% oxygen to patients during resuscitation. To ensure the risks of hyperoxia, we propose an oxygen bag that has a built-in oxygen blender that'll be able to titrate 100% oxygen in increments of 10% down to 21% Fraction of Inspired Oxygen (FiO₂): Vari-Ox bag. The Vari-Ox bag utilizes the pressure differential within the blender to passively suction in room air as the titration source. The pressure differential is caused by the input of 100% O₂ in the oxygen blender, which will mix with room air to deliver the correct amount of FiO₂ to the patient. Through having 3 slits open, each 1 cm x .5cm, and varying the inflow of 100% oxygen from 15L/m, 10L/m, 8L/m and 6 L/m, the titration level went from 74%, 66%, 62% then to 54% in the initial testing phase. With the successful creation of the Vari-Ox bag, a universal oxygen blender bag will be created to help battle both hyperoxia and hypoxia.



Department of Chemical Engineering

CHE.5 A Bridge to Renewable Energies: Carbon Dioxide Methanation

Ahmed Alghamdi, Joel Finlon, Michael Simon, Omar Takrouri

Environmental Engineering

Adviser(s): Dr. Betul Bilgin; David Bahr

Abstract: As the world shifts towards renewable energy sources to secure the environmental future of the planet, renewable energy industries have ran into the problem of energystorage. Because many renewable energy sources, including wind and solar energy, are not able to provide steady power generation throughout the day, storing renewable energy when it is overproduced so that it can be used when demand rises is crucial to the future of renewable energy technologies. The goal of this project is to design a plant located in Southern California that converts over-generated renewable energy to methane, thus chemically storing the energy to be used during electrical peak-use hours. The main driving economic factor for this project is the wide, predictable swings in electricity prices in California throughout the day. California energy prices regularly reach \$0 in the middle of the day when solar energy is readily available but then rise to become some of the highest in the country in the evenings when the sun sets. Storing free and low-price energy as methane allows the sale of cheap energy from midday solar generation on the high-priced evening market. An added benefit of this process is the capture of carbon dioxide that is used in the methanation process, which allows for a carbon cycle to develop between the gas power plant and the methanation process. As the state of California sets goals for zero-carbon energy across the state, this methanation process will integrate existing natural gas power plants into a renewable energy future and utilize existing infrastructure that could otherwise be abandoned in the changing energy industry.

CHE.8 Biodiesel Production from Soybean Oil

Dulce Colindres Mendoza, Yaser Khalid, Nisha Patel, Huda Usman, Advith Vashist

Bioprocess Engineering

Adviser(s): Dr. Betul Bilgin; Mike Armstrong, Technical Director, Advisian

Abstract: As the world population increases, there is a significant demand for the use of energy. The consumption of petroleum products such as fuels for transportation and polymers has led to the increase drilling and production of oil and natural gas across the world. Therefore, innovative method for generating more sustainable and renewable energy is essential to reduce the human dependence on fossil fuels. Biodiesel has been extensively studied as an alternate source of producing energy because of its renewable, biodegradable and non-toxic nature. Biodiesel is considered the most promising alternative fuel because it relies on a variety of combustibile renewable resources such as, animal fats, soybean oil, and/or reused cooking oil. The feedstock chosen for this project is refined soybean oil as it has a low amount of free fatty acids (0.5 %) compared to the other feedstocks and hence, demands little to no pretreatment. The catalyst for the transesterification stage of the plant was chosen to be sodium hydroxide along with methanol since it accelerates the rate of chemical reaction and lowers the reaction temperature. The goal of this project is to produce 49.78 million gallons per year which is 0.5 % of the predicted global production in 2020 i.e. 10.06 billion gallons. The plant will be in Peoria, IL as midwestern Illinois has the largest concentration of counties that produced 8 million bushels or more in the USA. The biodiesel produce through this process will be sold directly to refineries with a purity of 99.9%. The target gross profit of the plant is estimated to be \$96.38 million per year. The main product revenue to be obtained from the sale of the biodiesel is estimated to be \$196.05 million per year. The cash cost of production of the plant is calculated to be \$99.67 million per year. The Net present value after 15 years of plant life is predicted to be \$210 million per year.

CHE.4 Blue in the EU: Production of Blue Hydrogen for a Greener Future

Erica Caruso, Courteney Fabbri, Anne Serban, Maurice Smith

Petrochemical Engineering

Adviser(s): Dr. Betul Bilgin; Gavin Palmer, MEng, Brindley Engineering

Abstract: Most of the world's energy comes from oil and coal, both of which release large amounts of greenhouse gases into the atmosphere and cause detrimental effects to the environment. In order to decrease greenhouse gas emissions, many countries around the world like Germany have set strict standards in order to become extensively greenhouse gas-neutral by 2050. One way to implement this change is by targeting the transportation sector and converting fuel-guzzling buses, trains and cars to fuel cells. In order to power these fuel cells, hydrogen fuel and oxygen react spontaneously through a set of redox reactions, generating electricity as well as water and heat byproducts. Current methods for green hydrogen—which is produced by splitting water into hydrogen and oxygen through electrolysis—are very costly and cannot yet be competitive with the prices of petrol. Steam methane reforming, on the other hand, is a well-known process that takes pipeline methane and reacts it with steam to produce hydrogen, carbon monoxide, and carbon dioxide. By combining carbon capture with this process, environmental impacts are mitigated, and the final product is a consumer-friendly, high-purity blue hydrogen product that can be cost competitive with petrol. Given our chosen process of steam methane reforming (SMR), we plan to produce 1.5 KMTA of blue hydrogen in Weinheim, Germany to meet one third of the current demand for hydrogen fuel at H2 Mobility refueling stations throughout the country. Our SMR plant will be analyzed for its safety and process integrity as well as economic feasibility given the projected market. Our product will cater to the ramp-up of 250,000 fuel cell vehicles expected to hit the road by 2023 as well as present a cheaper alternative for clean energy to consumers. During the 20-year lifespan of our plant, we do not expect radical advancements in electrolysis technology to produce an affordable green hydrogen product that can compete with petrol. At around five times cheaper than green hydrogen, our blue product will serve as an important component in Germany's Climate Package to reduce carbon emissions by 55% from 1990 by the year 2030, as well as, to help Germany achieve fossil fuel independence by 2050.

CHE.9 From Raw Materials to High Demand Household Products-The Production of Polypropylene

Ahmad Abusubaih, Kathya Chavez, Jon Chiu, Yier Li, Mariam Murtaza

Petrochemical Engineering

Adviser(s): Dennis O'Brien

Abstract: Now more than ever, there is a worldwide demand for the hugest contributor in making household consumer products, propylene. Propylene is the main chemical product used in markets relating to packaging, textiles, plastics, technical products, and more. As the population is increasing 1.05% each year, the demand for propylene keeps increasing, leading to a 'propylene gap' between the demand and the supply of propylene. To fix the gap, a promising direction is to use the most well-known way of producing propylene, the propane dehydrogenation process (PDH). The technology that implements this process the best is UOP's Oleflex™ unit. It is a promising catalytic technology that can produce high purity and quality propylene from a propane rich stream- making this a leading process in successfully producing the primary product, propylene. Our goal has been to take this technology one step further. Our solution results in 99.7% product purity and 32% selectivity, creating a competitive product that's desired in the market. Additionally, based on today's prices, our advanced process is shown to be highly profitable, creating an NPV of 270 million dollars without including any profits from byproducts. Moreover, our process has significantly reduced environmental footprint of the amount of CO₂, NO_x, and NH₃ produced by a factor of 30%. Other aspects include the ability to have a fast startup (<36 months), higher on-stream availability, and an automatic catalyst regeneration system allowing for continuous operation. So even though the need for consumer goods is growing at a face pace rate, the process we designed inspired by the Oleflex™ unit aids in supplying the demand day by day, and ultimately providing the essential resources humans need now and in the future.

CHE.14 Human Waste to Biofuel

Gewargis Garmooos, Marvl Odeesh, Eduardo Salgado, Michael Stacy

Bioprocess Engineering

Adviser(s): Dr. Betul Bilgin; Jerome Palmer

Abstract: Fuel is one of the world's most sought out resources. Fossils fuels are a limited resource, and therefore alternatives are needed. According to BP Amoco, three major sources of fossil fuels exist: coal, natural gas, and oil, which are expected to last up until the year 2130. This problem leads to a high demand for renewable fuel. A great untapped renewable resource is biomass waste, and for the purpose of this project, human waste to biofuel. The Human Waste to Biofuel method will not only provide the potential of a renewable resource, but it will also provide an infrastructure that will contribute to public health, which leads to the goal of this project. For the purpose of this project Hydrothermal liquefaction method will be used. The rough economics from aspen yield an ROI of 13.80% and an IRR of 21.86%, this gives a 6-year payback period at 8% interest. Assumptions were made that the aspen values were low, and separate ISBL was assumed at \$30M. This gave a ROI of 8.94% an IRR of 17.40% and a payback period of 7-years. With so many public health concerns present in less developed countries, and human waste being one of the major components of these issues, the goal is to create a better tomorrow for these areas. With the removal of possible waste contamination of the water supply, the result should result in a decrease in the 24.33% infant mortality rate, and an increase in the 53.46% available drinking water in Lagos.

CHE.15 Hydraulic Fracturing's Dirty Secret

Evelyn Almonte, Emir Asani, Xuan Liu

Environmental Engineering

Adviser(s): Lance Baird, UOP; Dr. Betul Bilgin

Abstract: Hydraulic Fracturing continues to boost domestic oil production in the U.S. by accessing oil and gas in difficult-to-reach shale formations with horizontal drilling. Although this impact may seem beneficial, there are some negative factors that come with fracking. So, what is Hydraulic Fracturing's dirty secret? Fracking uses about 2 to 8 million gallons of water per well, while 3.4 million people die each year worldwide from scarce and contaminated water sources. Fracking has an average of 30% of flowback water returning to the surface. The most common ways of handling flowback water are deep-well injection and open-air pits. Deep-well injections can contaminate ground-water aquifers, while open-air pits evaporate harmful chemicals into the atmosphere. Consequently, treatment for flowback water is starting to become more common today. The typical process of handling flowback water is transporting the water to a municipal treatment plant for treatment and discharge. Our process takes this idea a step further; A mobile treatment solution. This solution will involve transporting our treatment system to fracking locations and treating the flowback water onsite. Our onsite, mobile treatment system enables treatment and recycling of fracking wastewater to help reduce the need to purchase fresh water, as well as reduce the environmental risk that storage of wastewater poses. Our goal is to treat about 55 wells a year that are located within the Marcellus Shale. Each site is guaranteed to have over 90% of their flowback water treated and ready for reuse. This service is the start of a revolutionary solution aimed to meet one of the world's greatest demands: clean water. Mobile treatment enables billions of gallons of water to be retained and reused, becoming an essential wastewater treatment vital for the future.

CHE.11 Para-xylene Production Through Methylation of Toluene

Mahamat Abdraman, James Fell, Rongwei Lin, William Rawson

Petrochemical Engineering

Adviser(s): Dr. Betul Bilgin; Adam Brostow, UOP Honeywell

Abstract: Paraxylene is a commonly produced product in the petrochemical industry. This valuable product is used in the making of polyester fibers, films and resins to satisfy the growing demand from emerging economies. The purpose of this project is to provide an innovative and novel design of a plant that will produce 99.5% pure paraxylene while maintaining the objectives of energy and cost efficiency. During the separation of paraxylene from a reformed naphtha feedstock, a large amount of by-products are produced, namely toluene. Toluene has a relatively cheap market price and low demand. Implementing the methylation of toluene process to the plant will be able to convert toluene into paraxylene with a conversion of greater than 90%. The design will implement an existing technological model of toluene methylation while also incorporating new energy optimization ideas that can still produce para-xylene of optimal purity. The plant is expected to have a production capacity of 4.5 MMPTA. The economic metrics justify the novel paraxylene production through toluene methylation compared to most conventional based and energy extensive processes. The process exhibits a high gross profit of \$1,092 MM/year and a cash cost of production of \$8,806 MM/year from a total revenue of \$3,158 MM/year. High production capacity and relatively low energy consumption exhibited by the process makes toluene methylation more appealing to investors in developing countries with high demand. The highly increasing consumption of paraxylene-derived materials by emerging economies, such as China and India, indicate that global demand might grow even higher than the current rate of 7% per year. Therefore, the optimum production of paraxylene through toluene methylation is lucrative and will remain as such for the next few decades as more underdeveloped countries move towards economic prosperity.

CHE.16 Plastics to Power: Producing Fuel

Zainab Ahmed, Sihang Chen, Ariel Maret, Allison Nordvall, Peter Starakiewicz

Environmental Engineering

Adviser(s): Patrick Shannon

Abstract: Due to the large demand for plastic, the consumption of various plastics has reached approximately 419 million tons in 2019. The World Bank reported that the proportion of plastic waste occupied in overall municipal solid waste all around the world has been nearly 8-12%. Additionally, China has announced a ban on importing plastic waste, which has led to changes in plastic waste management in the US. If not addressed, the abundance of plastic will cause issues for the environment and current recycling facilities. There are various methods to reuse plastic waste. Current methods include recycling into new plastic, fabric, and plastic roads. However, each has its own drawbacks. We have chosen to develop a plant that turns high-density polyethylene (HDPE) plastic waste into gasoline, a valuable commodity. The designed pilot plant uses 50 metric tons of HDPE and produces approximately 374 barrels of gasoline per day. In addition to plant design, this process has been analyzed from an environmental, safety, and economics perspective.

CHE.12 Polylactic Acid for our Bioplastic Assets

Diana Awimrin, Leena Daniel, Thomas Mazur, Yohannes Teweldeberhan

Environmental Engineering

Adviser(s): Dr. Betul Bilgin; Dennis O'Brien

Abstract: In 2016, 740 billion pounds of plastic was produced worldwide with an annual growth rate of approximately 4%. When considering that most of these plastics are oil-based, the drawbacks of using such plastics include an increase in waste management, land and water pollution, and the release of CO₂ into the atmosphere from the crude oil refinery process. With that said, bioplastics have seen a significant growth in interest compared to their oil-based plastic counter parts in the last several years. Bioplastics can be used in several important markets such as food packaging and beverage containers. The goal of this project is to design a production plant to produce polylactic acid (PLA), which is biodegradable and a key raw material in bioplastic production, while ensuring the plant operates as environmentally friendly and cost-effective as possible. PLA is projected to have an increased demand of 9% by 2022. It is imperative that any biobased plastics in consideration should be derived from plant material that does not require new land to be cultivated. This is because the cultivation of land to accommodate the growth of virgin feedstocks contribute greenhouse gas emissions. This plant will focus on producing PLA from the waste of an already-cultivated crop such as sugarcane bagasse. The plant will be located in São Paulo, Brazil producing 143 kT/yr of 99% purity PLA. The annual revenue is approximated to be \$244MM with a cash cost of production (CCOP) of \$166MM which leads to a gross profit \$78MM. If plastic producers adopt the production methods proposed in this report, greenhouse gases and waste will be reduced, while producing a sustainable and biodegradable product.

CHE.10 Powering India and Helping Africa's Economy through MTO Process

Sinan Masoud, Khalil Shehadeh, Emily Sobie, Esteban Vazquez-Elizondo

Petrochemical Engineering

Adviser(s): Bipin Vora

Abstract: The goal of the project is to increase Africa's GDP through investing in a Methanol plant due to the large reservoirs of natural gas in Angola. This will not only help boost their economy but will allow to provide India with lower cost methanol for the methanol-to-olefins (MTO) advanced plant. The project will only focus on the MTO process in India with the idea that India will invest in a methanol production plant in Angola. There is a growing demand of ethylene in India due to the population and economic growth they are experiencing. The demand will be satisfied with the MTO plant. Using the UOP/Hydro Process for the MTO process will be beneficial to India as the west-coast is filled with petrochemical plants. The goal is to either sell the excess power to plants in the surrounding region and to be able to control the ethylene and propylene selectivity through temperature changes of the reactor. We will use a SAPO-34 catalyst in order to get the highest yields out of our products. Furthermore, we will be able to optimize our utilities by reusing our products as refrigerants, which will in the end help with the optimization of our process. Plant capacity of ethylene and propylene is roughly 660,000 MT/yr.

CHE.3 Processing Green to be Green – Production of Renewable Diesel from Algae

Alice Guzik, Adisa Hadzic, Ian Lim, Alexis Miranda

Bioprocess Engineering

Adviser(s): Dr. Betul Bilgin; Lance Baird

Abstract: The world's increasing population is translating to an increase in the combustion of fossil fuel, thereby increasing the emission of carbon dioxide, a greenhouse gas. Globally, communities have been making strides toward reducing greenhouse gas emissions by switching to renewable fuel sources instead of using fossil fuels such as petroleum-derived products. This project uses green algae as a sustainable source of feedstock to produce renewable diesel. In comparison with other plant-based energy sources, green algae do not affect the human food chain because it is inedible and can be grown on land that cannot be used for crops; algae produce the highest yield of oil per acre because of its high growth rate and high lipid content. The renewable diesel produced from algae is a clean, stable, and high performing fuel that meets all specifications and has a high cetane number. The chemical composition of renewable diesel is comparable to that of petroleum diesel since, unlike biodiesel, it does not contain oxygen making it completely compatible with the existing diesel storage, transport, and dispensing infrastructure while retaining the ability to lower greenhouse gas emissions. Our project, inspired by UOP's Ecofining process, converts pretreated algae oil into renewable diesel using an integrated process of hydrodeoxygenation followed by isomerization. Our renewable diesel effectively reduces greenhouse gas emissions because it is produced from atmospheric CO₂ used by the algae to make our lipid feedstock. Based on a palatable oil market price for the lipid feedstock used in the process and a RIN value of \$0.50, the process is highly profitable, generating an NPV of \$159 million on an initial investment of \$223 million. Although there is a growing fleet of electric and hybrid cars, there will continue to be a need for renewable diesel for existing infrastructure as a low-cost, sustainable fuel that will reduce greenhouse gas emissions while equaling the long driving range and high performance of petroleum-derived diesel.

CHE.1 Production of Low Sulfur Gasoline using HDS

Fatimah Al Ibrahim, Javier Eduardo Mendieta Castillo, Christian Sanchez, Omar Zahdan

Petrochemical Engineering

Adviser(s): Dr. Betul Bilgin; Belma Demirel, BP

Abstract: The consumption of gasoline in the United States is relevant to the petrochemical industry since a considerable number of vehicles depend on this fuel. However, with the rising regulations to prevent environmental pollution from fossil fuel derived products, the sulfur composition of gasoline should be tracked and further treated to be lowered. As a result, the objective of this project is to design a hydrodesulfurization (HDS) unit in order to lower the sulfur content of gasoline produced via the fluid catalytic cracking unit (FCCU). The process includes adding pure hydrogen gas to the feedstream, which is FCC gasoline. In addition, the chosen catalyst CoMo/Al₂O₃ will be used along with hydrogen in the reactor to absorb the sulfur content and meet the new environmental regulations which require a sulfur content less than 10 ppm. The addition of the amine scrubber unit is then required in order to separate the recycled hydrogen stream from the hydrogen sulfide gas. When it comes to the control of the system, the main objective is to maintain and ensure the process is operating at the desired and optimal conditions for it to be safe and efficient. When designing the process, it is also important to consider the environmental concerns and the product specifications. The variables that we will be optimizing for our product specifications include, the low sulfur to meet EPA regulations and high-octane gasoline to meet customer demands after being post-treated using the HDS process. With the global population on the rise, the need for low sulfur high quality gasoline is required while ensuring lower air pollution. Oil demand is projected to climb by 8.7% with current production by the year 2030. With more gasoline required to be treated our process will ensure that the 40,000 BBL/day produced gasoline will be desulfurized in this plant.

CHE.6 Subdued Crude Paving the Way for Less-Lowd Renewed Car Food

Kellen Krupa, Milos Markovic, Jake Proctor, Cheryl Slykas

Bioprocess Engineering

Adviser(s): Jerome Palmer; Dr. Betul Bilgin

Abstract: Biodiesel is an essential product used in the fuel industry. It is typically blended with petroleum-diesel fuel for use in vehicles and is available nationwide in the U.S. It is also available as 100% biodiesel at many fueling stations across the country. The goal of this project is to design a chemical plant that produces biodiesel from used cooking oil via the supercritical processing method and to ensure cost effectiveness and energy efficiency. Traditionally, biodiesel is produced from vegetable oil using a catalyst to facilitate the reaction. By changing the process to the supercritical process, biodiesel can be produced without the need of a catalyst. This reduces the need for extra separation equipment and allows the production of technical grade glycerol as a byproduct. Market research indicates approximately a 7% growth of the biodiesel market in the U.S. over the next 6 years. It is estimated that this project fulfills approximately 1.5% of the available market which equates to 9.5 million gallons (226,190 barrels) of biodiesel per year.

CHE.2 Sweeten the Planet with Polylactic Acid (PLA)

Melissa Djonlich, Mariana St. Peter, Caleb Tristano, Jasmine Villegas

Environmental Engineering

Adviser(s): Dr. Betul Bilgin; Patrick Shannon

Abstract: One million plastic drinking bottles are purchased every minute around the world while more than 8.3 billion tons of plastic has been produced since the 1950's. Studies have shown that 60% of all plastic will end up either in landfills or natural environments. Petroleum based plastics can take thousands of years to degrade. The solution to reducing the environmental impact of non-degradable plastics is polylactic acid, or PLA: a biomass derived chain of lactic acid monomers that can be used to replace single use food-grade plastics. The FDA recognizes PLA as safe for all food packaging due to its non-toxic properties. PLA is also biodegradable, decomposing into biomass, methane, and hydrocarbons. The goal of this project is to design a plant in Shanghai, China that will produce PLA from glucose and optimize the design economically, sustainably and efficiently. Our process begins with the fermentation of crude lactic acid using *Lactobacillus* and glucose. Crude lactic acid then goes through a purification process involving reactive distillation and separation processes, resulting in purified lactic acid. Finally, the purified lactic acid is polymerized via a ring opening two-step catalytic reaction, producing around 150 tons/year of PLA, helping to mitigate plastic pollution around the globe.

CHE.13 The United States, an Energy Exporter

Tariq Hannoun, Diego Martinez, Erick Yspango

Petrochemical Engineering

Adviser(s): Dr. Betul Bilgin, Jon Prudhom, UOP Honeywell

Abstract: The outcome of recent unprecedented challenges to humanity that COVID-19 wrought will change the way energy is used, produced, and distributed globally. The rapid changes to the environment and fuel economy are already apparent, and it is possible that a valuable lesson will be learned from catastrophe. For now, oil's final hurrah will dominate the fuel market for the near future, at least until well exhaustion. According to current market data, there isn't much room for natural gas in the present, but the future is near. The design goal of this project is to purify Marcellus Shale natural gas to be liquefied and exported to Asia. Natural gas is abundant in the United States, and improvements in fracking technology had a positive impact on economic viability. Marcellus shale gas can be purified with the least effort with the intent for liquefaction when compared to western shale gas deposits due to its low concentration of acid gasses. Additionally, the Helium content of the natural gas feed is small enough that processing for Helium recovery is not required. The plant would be designed to process an amount of one billion cubic feet of shale gas per day. The mass flow rate of the final product stream approaches 99% of that of the feed. When all of the dust of current events settles, the demand for natural gas will return stronger than before. The United States needs to be able to attack that opportunity the moment it arises.

CHE.7 Tough Wrinkles? Bring in the Heavy Weight, Sodium Hyaluronate

Sandra Hernandez, Tianna Mitchell, Velia Palma, Yusuf Sorunke

Bioprocess Engineering

Adviser(s): Annette Johnston

Abstract: Hyaluronic acid (HA), also known as Sodium hyaluronate, is a popular ingredient in the skincare and cosmetics market using creams, serums and injections. Other applications include ophthalmic, biomedical, and dietary uses. The demand for hyaluronic acid is steadily increasing due to its anti-aging applications. The North America hyaluronic acid market is expected to reach a value of USD 41.3 billion by 2024 [3]. The primary goal of this process is to design a chemical plant in New Jersey that produces cosmetic grade hyaluronic acid. This will be accomplished through batch microbial fermentation. The chosen mutant bacterial strains of streptococcus A and C will yield 5-10 g/L of sodium hyaluronate with a molecular weight of about 1,800 Kilo Daltons, kDa from a sucrose feed. The plant operates batchwise, from fermentation to final product isolation. The fermentation growth cycle is about 20 hours, followed by a HA generation cycle of 7 hours. The recovery train was designed to generate packaged powder within the same week the fermentation is completed. The annual production rate of this chemical plant is of 32,500 kg with a total revenue of \$19.5 million and a gross profit of \$11.3 million. The plant development was analyzed from seed culture growth to packaging of the product. The economic analysis shows that financing this plant gives a feasible return on investment.



Department of Civil and Materials Engineering

CME.30 Adaptive Reuse of UIC's Former Chemical Engineering Building

Moshe Carron, Garrett Ott, Mateusz Pec

Site Development Engineering

Advisor(s): Dr. Didem Ozevin; William Loftus

Abstract: Adaptive reuse is the process of rehabilitating an existing building to serve a purpose other than was originally intended. Studies have shown that the construction industry alone contributes to approximately 20% of all air pollution and 40% of all drinking water pollution (Assefa 3). Ever-growing concerns relating to the impact the construction industry has on the environment have prompted designers and developers to step outside of the box with their ideas. Additional studies, conducted by Asseffa and Ambler, indicate that structural repurposing may significantly reduce the carbon emissions as compared to new build projects. Unfortunately, because of the nature of the design style, each project carries vastly different challenges both economic and engineering related. With this in mind, this senior design project will focus specifically on a case study addressing the challenges related to a structural repurposing of UIC's vacant Chemical Engineering building. Through the use of structural member testing and repair, as well as sustainable material sourcing, this project aims to create a path for other engineers to follow in order to reduce the impact our industry leaves on the environment.

CME.14 Analysis of Recycled HDPE Filament for Fused Filament Fabrication

Mikaela Brown, Francisco Canas, Justin Chambers

Sustainable Engineering

Advisor(s): Dr. Matthew Daly; Camden Treadway; Paige Kupfer, EI

Abstract: Fused Filament Fabrication (FFF) has been gaining popularity as a versatile option for prototyping and manufacturing. The most common filament materials include Acrylonitrile Butadiene Styrene (ABS) and Polylactic Acid (PLA), but there has been concern over the footprint and price from materials used during fabrication. In this analysis, previous research on the use of recycled High-Density Polyethylene (HDPE) as filament was explored to determine the suitability and sustainability of utilizing the material to design engineering-grade parts. HDPE is already used in a variety of applications including corrosive packaging, piping infrastructure, and consumer packaging making it a suitable candidate for consideration. Three main factors used to determine the effectiveness of recycling HDPE into filament include mechanical properties, economics, and environmental implications. It was found that not only is HDPE a suitable engineering material for recycling into filament, but the energy and environmental impacts of using recycled materials proved quite substantial.

CME.20 Application of Hybrid Roof Systems

Aimme Muro, Jesus Quintero

Sustainable Engineering

Advisor(s): Dr. Krishna Reddy; Joseph Bacha, EI

Abstract: For decades, green spaces across Chicago's industrial parks have declined, and many of the establishments that make up these parks demonstrate symptoms of deterioration. Hybrid roofing systems offer economic, environmental, and social support for outdated facilities in need of modernization. Focused primarily on an existing industrial warehouse located in Chicago, the proposal aims to design a thermoplastic polyolefin (TPO) roof integrated with an intensive green roof system. The report also explores case studies on intensive green roofs and their positive impact on thermal energy and water management. Additional research on key roofing elements are identified and critically reviewed. Software simulations provided by the Environmental Protection Agency (EPA) offer assessments of the roof's runoff rates, evaporation rates, and retention of precipitation levels. The design is heavily influenced by roofing contractors and vendors who provided consulting. Ten thousand square feet or one-third of the roof's surface area implements a green roof system while the rest practices TPO. Several findings reaffirm that medium-sized flora native to Colorado in a mineral-rich clay may be excellent candidates for the green roof. The green roof simulates as a viable rain garden and the National Stormwater Calculator estimates a retention of 16% of all annual rainfall. Energy assessments should utilize data provided by utility bills. The research concluded that wet and cold climates are susceptible to inaccurate data readings. National Roofing Company (NRC) provided essential hydrology tips and checked if the design was up to code, but one must note NRC are not certified inspectors. The project requires approximately \$800,000 and 23 working days. The proposal may eventually serve as a guide in implementing more intensive green roofs in the Chicago area, which can improve a variety of social, economic, and environmental aspects.

CME.18 Chicago South Side Land Development for Local Park

Abdel-Jabbar Alqam, Matthew Kurz, Jonathan Pillado

Site Development Engineering

Advisor(s): Dr. Joseph Schulenberg; John Miller

Abstract: In order to please residents in any city, measures must be taken to ensure their physical and mental well being. According to USpopulation.com, there are 2.75 million residents in the United States. Often times, residents do not have the time, means or even desire to go out of their way to spend time on recreational activities. Due to the recent success at Grissom Elementary School in Chicago, the project team will focus on the proposal and addition of a community center that contains an indoor center and a green park/playground at the corner of W 21st and S Jefferson St. This project will require engineering analysis including foundation design and stormwater management. According to the Chicago park district website, the Chicago park district budget was \$462,297,621 in 2018, with a total of 650 parks throughout the city. Despite that, many neighborhoods lack access to parks near them. Based on this information, the land development project will explore the possible additions and design limitations that are specific to this location. This will be done by determining the type, size and location of a foundation for the indoor portion of the center based on soil properties and characteristics. The water flow routing and stormwater management design will also be used to follow typical engineering practices. Sustainability is important for this design and an analysis of the most cost-effective materials will be conducted. Porous asphalt, bioretention landscapes or use of rain gardens are all possible green infrastructure components that could be implemented into the design. Visiting the site, determining the dimensions of the park, investigating the soil properties and determining the ground elevations are all part of the procedure to perform the analysis.

CME.3 Design of North Western Avenue Protected Bike Lane

Jeremy Borjas, Neftali Fonseca, Cintia Garcia

Transportation Engineering

Advisor(s): Dr. Jie Lin; Ryan Duffy, PE

Abstract: On a daily basis, numerous Chicagoans use bikes for recreational purposes and to run errands. Bike facilities allow individuals to travel from one destination to another in a safe manner. For some, bike infrastructure is a form of reliable and affordable transportation. For others it creates accidents, crime, and gentrification. Many roads in the city do not have a bike lane. North Western Avenue is a street located in Chicago near the neighborhoods of Ukrainian Village and Wicker Park. Currently, there is no dedicated bike lane attached to it. North Western Avenue hosts the end of bike lanes on West Armitage Avenue and West Augusta Boulevard intersections, as well as an entrance/exit ramp of the Bloomington trail; this gives it the potential to connect various bike routes. It is proposed that a protected bike lane be installed from West Armitage Avenue to West North Avenue. This proposed bike lane will help increase rider safety based on analysis from prior studies. AASHTO standards were followed, in order to design an adequate bike lane. The design was drafted by using Google Maps, MicroStation, and GIS. The design was then modified by inputting changes, based on various site visits. The expected results for the proposed bike lane were to decrease injuries of cyclists, increase the number of riders, decrease CO2 levels, and increase income of businesses in the area of interest. The results emphasize how creating protected bike lanes for the commuting public increases public safety, provides economic benefit for riders and provides a positive environmental impact by decreasing CO2 levels.

CME.25 Design of Pedestrian Bridges in Blue Island Illinois

Diego Alvarado, Cesar Dominguez, Ian Waratuke

Structural Engineering

Advisor(s): Dr. Mohsen Issa; Jeffrey Barnett, PE, SE

Abstract: Blue Island, Illinois is a suburb south of Chicago that is home to over 23,000 people with a number of businesses, train tracks, and schools. In particular, Dwight D. Eisenhower high school is located on the outskirts of Blue Island on 127th street and is separated by two tracks just east of the school. These tracks are frequently used and are almost exclusively used by freight companies. With the tracks being approximately 750 feet apart, walking commuters are bound to be stopped by one, if not both, of the tracks. The wait time for one track can be anywhere from 10-30 minutes. This greatly reduces the quality of life for residents of Blue Island, especially the students that choose to walk to school. To address this problem, one possible solution is to design two short pedestrian bridges. The design would also be ADA compliant. This would significantly increase the quality of life for the people of Blue Island. This project is something that the community desperately needs. This is due to the fact that students and other pedestrians have been seen crossing the tracks while in use. The pedestrian bridges will significantly reduce the possibility of someone being critically injured or losing their life.



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CME.7 Design Plan for Rehabilitation of the Damen Silos

George Lucaciu, Adam Palacio, Ismael Salas

Structural Engineering

Advisor(s): Dr. Mohsen Issa; Kaylee Knorr

Abstract: Chicago is a beautiful city with seemingly inexhaustible opportunities that promote daily exponential growth; and while that fairs well for the communities and businesses, it is wreaking havoc on the city's infrastructures. The State of Illinois received a grade of C- on it's 2018 American Society of Civil Engineers (ASCE) Report Card indicating that much of its infrastructure lies between "Mediocre: Requires Attention" and "Poor: At Risk". While the city's population thrives, the infrastructures are crippled by the increase in demand; structural deficiencies, such as section loss and formation of cracks on critical structural members, create multiple safety hazards and significantly decrease the overall service life. Proposed solutions for these problems are unappealing because the financial burden of demolition and reconstruction ultimately outweighs the end product benefits. As upcoming engineers and leaders we must adapt to the problems encountered; our team proposes a unique solution to structurally rehabilitate the Damen Silos. The Silos, designed and built by The John S. Metcalf Company in 1906 to support Chicago's grain trade, ceased operations due to an explosive accident in 1977. Located at the corner of South Damen Ave and West 29th Street, The Damen Silos currently serve as an urban adventure for those brave enough to bypass security. For our solution, we intend to structurally rehabilitate the Damen Silos with additional floor space to accommodate recreational, professional, and housing opportunities. Existing structural faults are to be mitigated while we choose stronger material for new structural members to compensate for existing conditions. We plan to offset the financial burden of reconstruction with use of environmentally friendly concepts, such as recycling the building material from nearby decrepit structures, while minimizing demolition. Additionally, the location is close to Chicago's downtown appealing to many housing and professional companies interested to use the space.

CME.10 Ductility in Pavement

Moe Husein, Hussein Mohd, Mahmoud Muhammad

Structural Engineering

Advisor(s): Dr. Mohsen Issa; David Dobson, PE; Saad Khatri, EI

Abstract: Traditional concrete has a low ductility. In a slab such as pavement, there is a higher probability of cracking when the bottom of the slab experiences a tension load. One example where this can be seen is when vehicles drive down a roadway. The goal of this project is to increase the ductility in pavement, making it more flexible, which will allow designers to reduce the bearing capacity of the soil in addition to this goal, the project resulted in a preferred mix design that would achieve the same or higher strength than the standard pavement mix by using a specified amount of fiber reinforcement. Laboratory experiments were done to determine the strength and ductility of three different mixes with varying amounts of fiber in them. The test results that were gathered will show how much the concrete in pavement will flex in order to apply it on a reduced bearing capacity. The result of the experiments was a highly ductile concrete which can be determined ductile and strong enough to allow pavement to be applied with a lower bearing capacity than traditional concrete. The research done showed that concrete that is heavily induced with fiber reinforcement would allow the concrete to flex enough to for it to be able to apply it to an area with a reduced bearing capacity. By doing this, it will decrease the soil removal costs on roadway projects because soil that would have normally been classified as unsuitable can now be left in place.

CME.12 Facilitating Wastewater Treatment for Irrigation and Agriculture in Illinois

Jonathan Diaz, Yuzhu Qiu, Raquel Rivera, Karina Rosiles

Sustainable Engineering

Advisor(s): Dr. Amid Khodadoust; David Klawitter; Elaine Avgoustakis

Abstract: Access to clean water for irrigation and agriculture has been a growing issue with climate change and the exponential need for more crops. In order to be able to meet these demands, some unconventional methods to reclaim water, include the reuse of greywater and treated wastewater. The pollutants found within domestic wastewater that may cause severe harm to the public include pathogens, heavy metals, Total Suspended Solids (TSS), and Synthetic Organic Compounds (SOCs). In order to prevent potable water from being wasted, reusing and treating wastewater from homes and buildings would save on expenses and would benefit the crops. This study focuses on the city of Urbana-Champaign, Illinois. The purpose of the water treatment design is to treat domestic wastewater into greywater and irrigation standards. The crops analyzed in this study were soy-beans. The crop will determine the nutrients kept within the treated wastewater. Wastewater treatment to agricultural and irrigation standards would reduce the expenses of farmers by limiting the external source of water to irrigate and using less fertilizers for crops. The wastewater treatment design in this report was successful in treating the wastewater contaminant levels to the acceptable greywater standards. All cases of treated wastewater are below the allowable turbidity levels and all pathogens and biosolids were captured. The domestic wastewater was put through four processes which included a granular media sand filter, a sedimentation tank with added carbon adsorption to remove the biosolids, and UV radiation to remove pathogens. The surge tank was designed as a storage facility for the greywater to be treated and pumped out to the appropriate irrigation systems. This process is less intensive than treating potable water and will reduce the energy consumption from 20-45 to 10-20 kW/PE*a and reduce CO₂ emissions from 36 kg CO₂/PE/a to 20-25 kg CO₂/PE/a.

CME.26 Green Infrastructure in the Town of Cicero, IL

Baihan Chen, Chiho Cheung, Peiyuan Jiang, Ran Tao

Water Resources Engineering

Advisor(s): Mike Burke, PE, CFM; Stephanie Maier, EI; Dr. Joseph Schulenberg

Abstract: Flooding is the most common and expensive problem in Illinois, particularly in dense urban areas. Our project focuses on stormwater management and flood reduction in the town of Cicero through the installation of green infrastructure. Stormwater in Cicero is collected in a combined sewer system, which drains both sanitary flow and stormwater runoff. The town experiences flooding and basement backups when the combined sewer system becomes overwhelmed after heavy rainfall. The purpose of green infrastructure is to decrease the volume of stormwater runoff entering the combined sewer system, which will lessen the burden on the sewer system. We chose a location within the town that currently produces a large amount of stormwater runoff. Drexel Elementary School was selected as the project location because it has a large amount of impervious area and in effect has a large amount of stormwater runoff. This project will utilize the Geographic Information System (GIS) software and EPA's National Stormwater Calculator to evaluate the effect of green infrastructure on reducing stormwater runoff. Four types of green infrastructure were evaluated, including: pervious pavement, vertical rain gardens, green alleys, and planter boxes. Based on this analysis, we will quantify the stormwater reduction benefits for the installation of these four types of green infrastructure.

CME.11 Green Skatepark: Bringing Flood Prevention and Urban Rejuvenation to an Underserved Community

Angelo Blanche, Ruth (Lu De) Ni, Chris Schaal

Sustainable Engineering

Advisor(s): Dr. Christopher Burke, PE; Oksana Zurawel

Abstract: Chicago neighborhoods are at risk of basement floodings and other costly damages from above average rainfall events because of poor infiltration and a lack of stormwater retention. Complicating factors include the lack of large parks and open spaces in many areas, which makes providing stormwater detention facilities without removing amenities a challenge. The goal is to battle the issue of urban flooding in the South Side area. For example, in the Parkway Gardens area on the South Side of Chicago there have been at least 84 NFIP (National Flood Insurance Program) claims made since 2000, totaling \$1.1 million. The low income population of this area, and the inordinate level of flood claims in this area compared to more affluent areas of the city, makes this an issue of environmental equity. In order to mitigate this issue, a stormwater detention facility has been designed. Using LiDAR data and municipal geographic information, a facility site was located that utilized natural terrain and maximized catchment area. Using StormCAD, SewerGEMS, and AutoCAD Civil 3D a detention pond and separate stormwater sewer system was designed and assessed for cost and flood protection. A detention basin within a local park with a skatepark and other green infrastructure has been strategically designed to respond to this concern. The desired outcome is to reduce the likelihood of flooding in the local area with stormwater detention and infiltration. Utilizing a novel design concept from Norway, the dry-bottom detention pond was designed to have a second purpose: that of a skate park for the enjoyment of local skateboarders. The idea has been presented to a local concrete design firm, Ozinga, to seek further development of a permeable concrete mix to fill in the non skating designated areas, by creating a detention pond that allows significant infiltration while still retaining its quality as a skate park. The design allows for the introduction of a unique park amenity in the South Side of Chicago, while providing stormwater detention consistent with Chicago's Stormwater Management Plan design criteria. This park's design has the ultimate goal to serve its local community, making a lasting yet positive impact by alleviating flooding.

CME.21 Halsted Parking Garage Addition

Alex DeFrancesco, John Isaac, Kyle Nerada

Structural Engineering

Advisor(s): Dr. Majid Mobasser, PE, SE

Abstract: In recent years the concept of mixed-use buildings has become the norm in dense urban areas. A mixed-use building is defined as a building with more than one purpose built into it. Examples of this can include a parking garage on the bottom of a building with residential units above it. In the case of our project, we want to retrofit the existing Halsted parking garage on the campus of the University of Illinois at Chicago with two additional stories on top of it to serve the role of a two-story study lounge. On the first floor of this new addition, we will be creating a modern computer lab and add additional tables for collaborative work. The second floor of this addition will be a floor entirely dedicated to private study rooms that can be rented out to students. This concept should be implemented as the undergraduate population at UIC has been rapidly increasing in recent years and additional study space would be extremely beneficial to future students attending this university. We also plan on adding additional safety measures to this building on the floor level of the parking garage due to past security issues at this location. For our results, we will provide a report on the projected cost of our project and the number of students that our additional space will be expected to accommodate. We will provide models of our proposed project along with drawing plans and renderings so we can fully demonstrate the vision we have for our project along with the specific details of what the inside will include.



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CME.9 Implementing Dry Wells and Recollection Units in Series to Remove Excess Runoff

Rogelio Argueta, Jose Gutierrez, Santiago Realmo

Sustainable Engineering

Advisor(s): Francisco Martinez; Dr. Joseph Schulenberg

Abstract: Drainage systems are important for the maintenance of structures that are subject to high volumes of precipitation. In Illinois, the annual precipitation levels have been increasing in recent times. Though this may not be a problem everywhere, it can become a major issue in places with drainage systems incapable of getting rid of excess rainwater. The Progridi group LLC is facing constant flooding on a gravel driveway, near a barn, due to the existing inefficient drainage system. Gravel driveways may be able to deal with the direct runoff of rain on its own. However, in this case the accumulated water exceeds the draining capacity of the gravel. There exists a plethora of solutions to deal with excess runoff, including the use of dry wells. A dry well is a subsurface storage system that temporarily stores stormwater runoff and dissipates the runoff into the surrounding soil over time. The team chose the implementation of a dry well to accomplish a reduction of runoff accumulated in undesired locations at the site. In order to obtain better results, an extreme precipitation scenario was assumed to begin the design process. The dry well design was based on the infiltration rate of the surrounding soils, and the location was designated according to the existing construction requirements and specifications. Although the dry well decreases the amount of rain water discharge, the team also considered implementing a filtration system to recollect a portion of the excess runoff. The incorporation of these additions to the existing roof drainage system to prevent the risk of flooding in the driveway.

CME.27 Integration of Geothermal Heat Pumps in a Building Foundation

Cody Cetnarowski, Ashley Griffin, Karina Kuyek

Sustainable Engineering

Advisor(s): Dr. Krishna Reddy; Richard Jezierny, EI

Abstract: With the increased concern of climate change, green structures are becoming more popular and in demand than ever. In fact, according to World Green Building Trends 2018 SmartMarket Report, more than half of the survey respondents are expected to increase their green projects to 60% in 2021. Since there is an increased focus on sustainable solutions, engineers are able to improve or create new and unique designs for future infrastructure to meet these demands. An integrated geothermal heat pump system in a building foundation can be a cost effective, safe, and clean alternative to conventional energy consumption. The University of Illinois at Chicago (UIC), as an example, has installed fourteen geothermal wells in a nearby field to heat one of its buildings (Grant Hall) to keep it at a consistent 73°F year round. Designs like this allow for an advancement of simplistic and practical applications of geothermal energy in structures. The objective of this design is to integrate geothermal energy caissons into the foundation structure of the new Computing, Design, Learning, Research Center (CDLRC) at UIC that is scheduled to break ground in 2021. The geothermal energy pipes will be integrated into the caissons to support the structure and maximize thermal energy output to assist with the heating of the CDLRC to save in energy consumption and CO2 emissions. The expected results include an analysis of reduced CO2 emissions due to heating and cooling up to 90%, as well as a financial analysis which, has the potential to save up to 50% of the energy costs compared to a typical cooling system. It is expected that the pipes will last approximately 50 years before they need repair, so there will also be an expected return on investment for the overall geothermal system.



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CME.17 Kimball Avenue Brownfield Site Remediation, Redevelopment, and Attachment to the Bloomingdale Trail

Faris Eisouh, Martin Figueroa, Hashim Suleman, Manuel Velasco Velasco

Site Development Engineering

Advisor(s): Dr. Hossein Ataei; Shawn Murphy, PE

Abstract: This project will focus on the remediation and redevelopment of the 1807-1815 North Kimball Avenue brownfield site, followed by attachment to the 606 Bloomingdale Trail. A brownfield site is defined by the presence or potential presence of a hazardous substance, pollutant, or contaminant. According to an environmental assessment performed by the City of Chicago, the Kimball Ave site has been subjected to decades of pollution and contamination from chemical waste products that leached into the soil from the various industrial and manufacturing facilities that have been located on the site, dating back to the late 1800's. Land use and development can significantly impact the social, emotional, and economic well-being of a community and its residents. This project will rehabilitate and redevelop a prime plot of land that has been destroyed by decades of pollution, and improve it for public use to enhance community health, create new recreational space, and improve social equity. Remediation will be achieved by first removing and replacing soils with the highest concentration of harmful chemicals, followed by in-situ chemical treatment of remaining contaminated soils, and finished with driving of sheet piles around the site perimeter to anchor the soil in place and prevent harmful chemicals from polluting neighboring sites through groundwater transport. Redevelopment will include regrading and installation of Rain Gardens and Porous Concrete to address rainfall runoff within site boundaries, followed by paving of the 606 Bloomingdale Trail attachment, and installation of additional recreational attractions. This project will deliver a multi-benefit public greenspace serving the surrounding communities of Logan Square and Humboldt Park.

CME.6 LEED Certified Green Roof

Aya Aladhami, Anna Dudek, Yunfei Ma, Anisha Patel, Jianxin Zheng

Sustainable Engineering

Advisor(s): Sarah Koning; Nicholas Haas, EIT; Dr. Amid Khodadoust

Abstract: Green Roof designs provide a multitude of benefits to a building and its surrounding areas, especially for buildings within the city of Chicago. Chicago is considered a nonattainment area for ozone, therefore the city has taken action to change that. The city has taken serious steps to incorporate green roofs onto various buildings in order to reduce the urban heat island effect, a phenomenon where the city holds more heat than its surrounding suburban areas. This is due to the fact that concrete buildings give off excess heat, as well as anthropogenic heat production increases, raising the temperature of the air. Green roofs may be placed on any type of building in order to save energy costs. On an average, the city of Chicago receives about 36.89 inches of rain per year based on annual average rainfall between years 1871-2019. A study has been done on the Chicago City Hall green roof, and it was shown that due to the installation of the green roof, about 9,000 kilowatt-hours of energy per year were saved. The Chicago City Hall green roof had also shown to reduce the load on the stormwater sewer system. Based on the research that was done, a green roof design for the University of Illinois Hospital and Health Science System Outpatient Surgery Center and Specialty Clinics building will be a great addition for the campus. The purpose of this green roof is to absorb and eventually capture excess rainwater for internal usage, by means of flushing toilets in the building. This design will save a large amount of water usage, in turn, saving costs for water. Various other benefits will be provided due to the installation of this green roof such as reduced building energy costs, improved air quality, pollution reduction, as well as helping to reduce the urban heat island effect. This project will be unique to University of Illinois at Chicago because it will be the first green roof that is being designed for that area of campus. This project will also have an added educational value, for students and faculty in understanding all the benefits that come with a green roof design, as well as learning how to maintain it. This roof design will be a great way to make use of natural resources, thus improving the urban environment that is Chicago. This will help spread awareness of the importance of sustainable living.



Engineering Expo 2020

CME.1 Measure of Environmental Impact Based On Traffic Congestion

Daniel Koltys, Philip Kwasinski, Matthew Lebieczinski

Transportation Engineering

Advisor(s): Dr. Jie Lin; Joseph A. Lewis, PE

Abstract: Illinois Route 47 between U.S. Route 14 and Charles Road is predicted to endure severe amounts of congestion over the next 20 years leading to a substantial increase in emissions. Although current congestion levels are moderate, if untouched this will cause serious traffic impacts with environmental side effects for future citizens of Woodstock (Illinois, USA). Typically, passenger vehicles emit around 4.6 metric tons of carbon dioxide per year. This can be reduced since carbon dioxide emissions increase if a vehicle is going at a speed of less than 45 mph. These issues affect the longevity of the planet which is why a plan was devised to target traffic congestion, and will promote positive emission ratings. Woodstock consists of major routes in the surrounding area such as IL Route 47 and U.S. Route 14 where most congestion occurs. To combat this, a plan was devised to analyze the congestion levels within these routes. Three cases will be modeled: previous road conditions, future road conditions with Illinois Department of Transportation (IDOT) improvements, and future road conditions without any modifications. These models will be created and analyzed for three main objectives. These objectives are congestion with emission reduction, usability of the newly designed structures, and the economic feasibility of the project long term. IDOT design reports suggest that the five mile strip will eventually decrease in traffic conditions from moderate to acute throughout various areas. Previous design work showed that widening the roadway increased the level of service. This change shows that congestion can be reduced, thus increasing the average speed over 45 mph. Applying this approach with IL Route 47 model will show a positive change in level of service as well as delay times, density, and rush hour speeds through reducing greenhouse gases and pollutants from route modifications.

CME.4 Metra Regional Express Rail

Kyle Middleton, Jawad Rahman, Evan Storm, Daniel Wojcik

Transportation Engineering

Advisor(s): Dr. Farid Peiravian; John LaPaglia, PE; Brian Kubilius, PE, CFM

Abstract: Chicago Union Station is the primary terminal for all Metra and Amtrak passenger trains operating in and out of the City of Chicago, including both intercity and commuter trains. This project will analyze Metra's opportunity to offer cross-town services that pass through Union Station instead of terminating there. With the addition of more frequent service, Metra's operations will be considerably more attractive to both existing customers and untapped commuter markets, who frequently make long drives between suburbs. The objective of this project is to demonstrate how this operating scheme could be affordably implemented, in multiple phases, on the Metra system while using as much existing infrastructure as possible. Metra Regional Express Rail will provide residents with the opportunity to take Metra to and from surrounding suburban areas with the efficiency of rapid transit through Union Station using new larger platforms and open concourse designed for safety and ease of boarding. The results are expected to show a significant increase of residents commuting on Metra throughout the Chicagoland area and surrounding suburbs by improving the overall service area, encouraging a greater number of residents to utilize the transit system. Residents will be satisfied with the expanded transportation options offered by Metra, with higher operating frequency enabling commuters to get to their destinations more reliably. A wider selection of destinations on selected lines will allow Metra to break into new markets and compete with less environmentally friendly modes of transportation while opening faster travel opportunities for those looking to travel through the city and not necessarily into it.

CME.13 Multi-Hazard Vulnerability Study of Chicago Homes: A Case Study from Englewood

Yu Dong, Caitlin Koziel, Nalin Naranjo, Tanja Rakovic, Chenxi Xu

Structural Engineering

Advisor(s): Dr. Didem Ozevin; Dr. Craig Foster; Joseph DeFrenza

Abstract: For years, Englewood residents have been struggling to maintain the structural integrity of their homes. This study determines how that structural integrity is impacted by settlement, flooding, aging infrastructure and vibrations caused by cargo disembarkment at the CSX Intermodal 59th Street Yard. Residents' quality of life is revealed by analyzing discomforts that come with noise pollution and lower magnitude vibrations. Residents claim that cargo disembarkment from trains causes excessive vibrations which contribute to structural damage and exacerbating flooding issues. To investigate these claims, this study analyzes vibration and thermal data from a participating Englewood home, as well as soil and weather data from the area. While general studies have explored the impact of vibrations from passing trains or mine blasts on structures, there has not been a study that determines cargo disembarkment's impact on nearby structures. The Englewood study is unique as it analyzes the effects of vibrations caused by freight rail yard activities and how they contribute to structural damage to the common Chicago Worker Cottage. In order to confirm that vibrations are significantly impacting home structures, vibrations must exceed a PPV of 0.1 in/sec. To combat structural damage and discomfort, mitigation designs were developed. Using SAP and ANSYS models, the impact of settlement, potentially the main source of structural damage, is evaluated. Appropriate design solutions include modifying the structure's vibration characteristics by strengthening the slab, columns and foundation, minimizing water intrusion in susceptible areas identified by an infrared camera and comparing stresses due to support settlement for wood column and masonry wall structural systems. The mitigation strategy is implemented into the finite element models to demonstrate how structural damage and vibration levels can be reduced. These potential solutions can later be supported by additional measurements and implemented into residents' homes, thus improving their quality of life.

CME.22 Paving the Forward with Recycling Plastic

Jim Brennan, Arthur Dlugi, Maciej Orlowski

Sustainable Engineering

Advisor(s): Dr. Matthew Daly; Nicholas Haas, EIT

Abstract: In recent years plastics have become the most common material used for about everything, from packages, bottles, cups, and many more products. It was measured that there are around 275 million tons of Global plastic waste per year, and that's not including the 8 million tons of plastic in the oceans. Joe Iosbaker, Recycling Coordinator at UIC stated, UIC alone is "collecting about 3 tons of recycled waste a month from the buildings across campus. 90% of that by volume is plastic". However, plastic can be used again by forming it into pavers. Management of recycled plastics as material for pavers might help reduce the problem and implement the alternative for recycling. Polyethylene is one of the most commonly recycled plastics and can be reused by application of producing plastic pavers. When reusing most types of plastic materials, we can gain less expensive material of poorer quality. This material will not have the same mechanical characteristics as the original plastic. However, obtaining materials with the appropriate characteristics can be possible. The process of shriveling and melting of plastic material selected must be adequate to the result that will not take away much of material strength, sustainability and molding ability. This project investigates using multiple different thermoplastics to form pavers, by producing a method to shred the plastic, melt it, and form it into pavers. These pavers are tested to determine their strength and are compared to conventional pavers. It is envisioned that these pavers can be used in a myriad of solutions but are primarily intended for pavement applications. Ultimately, this project's research and analysis has indicated that recycled plastic pavers are a viable product that will reduce the amount of plastics thrown into landfills by providing an alternative to reuse plastic.

CME.19 Re-Routing Excess Flood Water of Buffalo Creek to the Des Plaines River

Cyan Brown, Mayyada Donley, Niko Njaastad

Water Resources Engineering

Advisor(s): Joseph Wilk, EI; Jeana Gowin, PE, CFM; Dr. Joseph Schulenberg

Abstract: Buffalo Creek within Wheeling, Illinois presents an ongoing flooding problem. The village's situation is similar to that of other communities along the Des Plaines River. Living along Buffalo Creek can prove to not only be a dangerous endeavor, but a costly one as well. According to a Detailed Watershed Plan done by Christopher B Burke Engineering, approximately \$68 million in existing condition damages have been endured over the past 50 years. Yet, despite the implementation of other flood mitigation projects around the stream of water, Buffalo Creek still shows many locations in need of change. The stormwater diversion tunnel that was created in Albany Park provided a great method of decreasing urban runoff; this design aims to serve as a similar solution to the Buffalo Creek area. There are multiple ways to alleviate the area's flooding problems, but the use of an underground stormwater diversion pipe shows to have the most promise for the community in the long run. Of course, Wheeling, IL isn't just an open stretch of land with the feasibility of any engineering project. Researching the area has shown that many places within Wheeling simply do not have the required quality nor quantity of the available space. This project will be about pinpointing an area in need of flood control and considering the feasibility of an underground flood pipe. In order to ensure the feasibility of the pipe, a couple of objectives will be considered: reducing the effects of flooding due to Buffalo Creek in Wheeling, IL and considering the impact of construction/project implementation on the environment. With this pipe, the expected results should be a decrease in homes flooded as well as a reduction in the depth of ponding in the area to 6-inches or less.

CME.15 Site Inspection using UASs

Majd Masri, Muntather Merei, Adam Saadi

Robotics Engineering

Advisor(s): Dr. Mohsed Issa; Edmund Burke

Abstract: Unmanned Aerial System (UAS) is an application of using aircraft, such as drones, to do a site inspection in construction sites. The goal of using a UAS is to collect data, such as images, to analyze them and monitor ongoing operations for inspection purposes in construction sites. The benefits of using UASs is to minimize the need for the labor force, ensure the safety of the environment, minimize the amount of time to complete a task, and ensure the safety of people inside and outside construction sites. As shown in previous applications, the accuracy of data collection, through UASs, has developed over the past years to be used by contractors. In this project, the team will use a drone to do a site inspection of a land that could be used for future projects. Inspection includes surveying, topographic mapping, damage assessment, and 3D modeling to observe the land use. The drone that will be most useful to this project is DJI Phantom 3 and Drone Deploy software will be used to fly the drone. The drone will be used for taking pictures of the land. Then, data will be analyzed by using Pix4D mapper software to generate a 3D model. According to a recent study by Stöcker, C., Bennett, R, Nex, F., Gerke, M., & Zevenbergen, J. (2017) "Although UAVs have their origins in military contexts, they have also become valuable for scientific and commercial applications, particularly over the previous decade. Various stakeholders and actors, including governmental authorities, commercial operators, scientific institutions and individuals, make use of UAVs as an affordable data acquisition tool: they enable mapping at temporal and spatial scales that still remain unachievable for traditional remote sensing platforms." The study indicates that those visual inspections of large sites are time-consuming and risky before using drones but nowadays drones are needed most in the industry because they are an important matter in construction sites and that is why engineers take the use of UASs seriously.



CME.28 Sustainability Design of Multi-Purpose Building in the 27th ward

Zayed Kiswani, Jacob Miguést, Gabrielle Yates, Christian Zumba

Structural Engineering

Advisor(s): Sasha Simon, El; David Klawitter; Dr. Sheng-Wei Chi

Abstract: The Cabrini Green area in Chicago was infamous for low-income housing projects. In 1994, the Chicago Housing Authority (CHA) planned to transition Cabrini Green from a low-income community to a mixed-income community. The area has since been completely regenerated and gentrified with high price condos, including multi-family and single-family homes. Even though the strategy reduced unsafe conditions, and created a healthier neighborhood, there is still a need for mixed-income units in order to prevent the exclusion of low-income residents from the community. Additional commercial and residential spaces will be developed in the area and the project design focus on the sustainability of these spaces. The integration of sustainable mixed-use structures, coupled with the functionality of the establishments they house, can add value to the community. This project will develop a mixed-use commercial apartment building to satisfy the needs of the community. Considerations for the life cycle of the structure and materials were made in the design process, including resource depletion and material toxicity. Chicago building codes are taken into consideration regarding the loads that must be accounted for in the design. The structure will be analyzed by modeling and analysis software. The software will also showcase the layout of the apartment. The design process includes how the structure interacts with people and the natural environment. The 20,000SF structure will consist of a 10 floor, 144-unit apartment tower which incorporates an innovation center on the first floor. 10% of the available apartments were designated to low income units. The project aims to reduce carbon intensive materials and energy costs while filling the housing needs of the community.

CME.2 Sustainable Construction Methods for 'Industrial Park' Development in Chicago's Southeast Side

Gilbert Arceo, Jose Mendez, Steven Rico

Site Development Engineering

Advisor(s): Dr. Joseph Schulenberg; Andrea M. Maertens-Pizzo, PE, CFM; Dena Delpazir

Abstract: With the higher demand for sustainability, the project will focus on the implementation of sustainable methods on the newly proposed "Ave O Industrial Park", a development site by Chicago's southeast side. The construction industry accounts for a large percentage of carbon emissions. As stated by the United States Environmental Protection Agency (EPA), buildings alone account for 39% of the total energy consumed in the United States and for 38% of carbon dioxide emissions. For these reasons, there is an immense need for sustainable construction methodology to be applied in the preliminary planning and delivery methods for construction development. Thus, sustainable courses of action during the pre-construction and construction phases will focus on achieving a decrease in waste production for "Ave O Industrial Park". This will be done through the minimization of material orders and the reutilization of on-site materials, such as the use of crushed asphalt for a more sustainable sub-base filling. Additionally, careful selection of materials is investigated and analyzed to highlight the benefits of lowering the amount of carbon emissions. Reducing the overall carbon footprint up to 30% will be achieved through the selection of materials, such as higher percentages of fly ash in concrete and the use of recycled steel or light-gauge steel frames. The project will also compare traditional construction methods and sustainable construction methods, as well as using a modeling tool such as ArgonneGreet.NET to compute energy consumption and greenhouse gas emissions. As a result, these techniques will be highly recommended for the completion of the industrial park.

CME.5 Taylor Street Innovation Using Complete Streets

Eduardo Alcorchas, Alex Blizek, Ricardo Vega

Transportation Engineering

Advisor(s): David Kleinwachter, PE; Dr. Farid Peiravian

Abstract: A Complete Street is a unique roadway that is designed to be safe and accessible for all users within that roadway which include pedestrians, bicyclists, public transportation users, and motorists. Complete Streets is defined under Illinois Public Act 095-0665 as a law that directs the state to establish pedestrian and bicycle accommodations in the planning and reconstruction of all state road projects. In Chicago, there are various streets that have not been redesigned or up to date with the latest specifications and needs for our growing population. While constant work is seen on paving the roadways and applying new concrete sidewalks, curbs, and gutters, there is still more design work to be done other than routine maintenance in order to have "Complete Streets". Many Chicago streets lack dedicated bicycle and bus lanes as well as accommodations for public transportation users. The existing design of streets accommodates mainly drivers and does not take into consideration the growing number of people using alternative means of transportation. The scope of this project will be dedicated towards updating an insufficient roadway in the city in desperate need of new pedestrian and bicycle accommodations. The street chosen for the project will be Taylor Street from Halsted to Morgan Streets. Complete Streets will include updated sidewalks with emergency booths for people walking at night, dedicated bike lanes, and dedicated bus lanes with accommodations for those waiting at the bus stop in order to achieve a more convenient and safer commute for users. This street was chosen because many students use it everyday to walk from various buildings and facilities around the UIC campus. Taylor Street will be enhanced using CDOT specifications in order and to bring it into conformance with the Illinois Complete Streets policy and create safer accommodations for all users of the roadway.

CME.24 Thermoelectric Pavement Heating and Generation System

Austin Meartz, Josh Polek, Artem Vyshnyakov

Sustainable Engineering

Advisor(s): Douglas Kerr, PE; David McDonald

Abstract: As part of the UIC Capstone Project, sustainable energy harvesting and pavement heating via thermoelectric generation were examined. The objective of the study was to create a technology to prevent snow and ice accumulations on roadways, airport runways, and sidewalks, while significantly lowering energy consumption and providing an additional source of renewable energy. The proposed technology has been termed Thermoelectric Pavement Heating and Generation Systems (TPHGS). The technology utilizes thermoelectric generators (TEGs) that are capable of harvesting thermal energy and converting it to electrical energy when subjected to different thermal environments. Moreover, TEGs can receive electric current and be turned into heating elements. A single prototype was created, consisting of two copper plates shaped as triangular prisms; one of which was made smaller to fit inside the other. Between the plates were installed TEG units that would use the temperature differential of the plates to generate electricity. Theoretically, the bottom end of the TPHGS would be installed in the subgrade (soil) of a pavement section, and the TPHGS would be installed in the concrete slab of the pavement section. Due to the COVID-19 outbreak and campus restrictions, the team was unable to make a pavement section and perform applicable testing. However, it was determined that when the TEGs were supplied with 90 Watts for 15 seconds (0.375 W-hr) the temperature of the outside copper prism was raised by 30°F, which would successfully prevent snow and ice accumulation in most applications. Based on TEG manufacturer's data a single prototype would also be able to generate over 3.3 Watts of electricity when provided with a temperature differential of 36°.



Engineering Expo 2020

CME.16 Updated Bulletin 70 Evaluation, Stormwater Storage Design

Layan Abu Than, Devonte Evans, Peter Morris, Mark Regalado

Water Resources Engineering

Advisor(s): Dr. Christopher B. Burke, PE; Luke Sherry, PE, CFM

Abstract: Within the past decade, the region of Northeastern Illinois has experienced more frequent severe storm events that have resulted in damages of approximately \$1.5 billion dollars. Typically, the rainfall depths and intensities exceeded the design return period values of the Illinois standard for extreme rainfall events, Bulletin 70. In response, an update to Bulletin 70 was developed in 2019 by the Illinois State Water Survey, resulting in increased rainfall depths and updated rainfall distributions. This is the first update to Bulletin 70 since the standard's original development in 1989, and the updated standard has been adopted by regulatory agencies to go into effect in 2020. A hydrologic and hydraulic analysis of the McDonald Creek Watershed, which is located in the Northwestern Chicago suburbs, will be evaluated using the updated Bulletin 70 depths. The analysis will evaluate the riverine system, as well as the Lake Arlington Flood Control Reservoir, to determine the effects of the new rainfall depths on the flowrates, flood elevations, and reservoir performance of the watershed. A hydrologic and hydraulic analysis of the McDonald Creek Watershed was completed in 2010 using the 1989 Bulletin 70 rainfall. That modeling will be updated using the 2019 Bulletin 70 rainfall depths and distributions so that the performance of the system with the new standard can be determined. The US Army Corps of Engineers' (USACE) HEC-HMS software will be used for hydrologic modeling of the watershed, and the output of the hydrologic model will be used as inputs to the USACE HEC-RAS hydraulic model to determine the stream and storage response. ArcGIS software will be used to map the results of the updated analysis. A design for improved stormwater storage will be produced adhering to the updated standard. Both online and offline storage schemes will be evaluated to determine the most suitable design alternative to accommodate the increase in flowrates and flood elevations. Conceptual construction cost estimates will be developed for those storage alternatives.

CME.29 Use of Adaptive Traffic Management System to Reduce Queue Spillback

James Berggren, Michelle Calcagno, Tresor Moolo, Yuna Wang

Transportation Engineering

Advisor(s): Bobby Gunnells, PE; Dr. Farid Peiravian

Abstract: Queue spillback is a major contributor to congestion in heavily traversed urban intersections. Drivers who queue in an intersection block cross movements, completely restricting traffic flow and greatly reducing intersection and system efficiency. Traffic congestion does not only cause delay, but also negatively impacts the environment, economy, productivity, and quality of life. The cost of congestion in the City of Chicago, alone, is over \$7 billion annually. Studies performed using computer models and connected vehicles indicate that improved signal coordination and adaptive traffic management systems are effective methods for improving traffic flow in a system and reducing queue spillback. Additional models have shown that the improved traffic signal effectiveness from real-time traffic detection significantly reduces congestion and emissions. This study seeks to demonstrate to a local department of transportation the benefit of implementing a real-time traffic detection system in a test area bordering the Loop in downtown Chicago. The primary objective is to prevent the spillback that causes congestion by utilizing an adaptive traffic management system. The second objective is to demonstrate the benefits of bringing Chicago up to date technologically as well as improving emergency management capabilities and communications with other municipalities. This is to be achieved with the use of Video Image Processors (VIP), included in the adaptive traffic management system, that can be used to identify and fine motorists that block intersections. The study will compare the current system traffic flow to the traffic flow in an adaptive traffic management system in the study area to determine the impact on the Level of Service (LOS). LOS analysis will be used to estimate the changes in emissions, fossil fuel consumption, and the cost of congestion.



Department of Computer Science

Honors Capstone Project AAll Member Behavior Predictions

Jonathan Uhlmann

Abstract: The American Association of Individual Investors (AAll) is a nonprofit corporation with the purpose of assisting individuals to build their investment wealth through programs of education, publications, and software. Using customer order and activity data from AAll, a machine learning model will be able to accurately predict the members' next actions in the near future. The goal of this project focuses on predicting member lifetime value, and next purchase day. The resulting data can be used by AAll to optimize their marketing strategy, and target certain member groups that will generate more revenue.

Honors Capstone Project Crowdsourcing Wildlife Data From Social Media

Ellen Kidane

Abstract: With modern image analysis technologies and platforms like Wildbook, it is becoming possible to use images from social media platforms to identify individual animals and obtain knowledge about endangered species populations. In this study, we collected images of whale sharks, humpback whales, reticulated giraffes, iberian lynx, Grevy's zebras, and plains zebras from YouTube, Flickr, Twitter, and iNaturalist. Our data showed that the number of posts, the frequency of total wild encounters, and the posts per user varies across platforms and species. Nonetheless, social media platforms present the opportunity to be a source of wildlife monitoring data for future risk assessment.

CS.5 Arduino Blackjack

Tarun Maddineni, Aditya Vandanapu

Abstract: This project aims to transform the popular game of blackjack into a two-player game between Arduinos. Each Arduino will have buttons that allow players input whether they want a card dealt or if they are done drawing. They will take turns drawing more cards until they are unable to or don't want to. There will be a screen keep tracking of the score between the players. Once a player has reached a winning score the Arduinos will reset the game automatically after a short period of time. This will allow for multiple games to be played between the players.

CS.18 Arp Cube

Wisam Abunimeh, Hanna Thayyi

Abstract: The ArpCube is an Arduino powered MIDI controller. The cube can be plugged into a computer to process the button controls into MIDI. The ArpCube can act as a standalone MIDI trigger or it can be used as an arpeggiator. An arpeggio is a musical term for playing the notes of a chord in succession. The ArpCube allows a user to step through a selected sequence of notes based on the input buttons. The ArpCube does not support MIDI-IN or MIDI-THRU signals, only MIDI-OUT.

CS.14 Automated Chess

Julian Gonzales, Daniel Hrubec

Abstract: Automated Chess using Arduinos. The Arduino will take in input from the user on an input square and output square. Then using Serial Communication, it will communicate the user input to a second Arduino. This Arduino will then read in the input, being the starting square and the target square. Using stepper motors, it will move the piece from the starting square to the target square.

CS.43 Automated Watering Gun

Jay Patel, Kunj Patel, Raj Prajapati

Abstract: We will be making an automatic water gun to water the plants by using a timer based mechanism. The target audience for our product would be the people with gardens in their house, commercial buildings or Schools with gardens. Our Aim is to help people make hassle-free travel plans, without worrying about watering their garden. If it is being used at Schools or Public Gardens, it will reduce labor.

CS.8 Battleship

Jabriel Crawford, Alex Vanberkum

Abstract: We are going to make a battleship game with arduinos. The arduinos will use lights, buttons, and Bluetooth communication to make the battleship game into Arduino form.

CS.45 Biometric and RFID tag Security System

Vatsal Patel, Yash Shah

Abstract: The Biometric and RFID tag security systems are aimed towards hotels and multinational companies where only the employees can be allowed to enter a specific area. Firstly, the employee must scan the RFID tag and then they can proceed for their biometric scan whose fingerprint will already be stored in the system for access. This ensures that only the employees of that specific company can enter a restricted area

CS.15 Bluetooth Remote Control Car with Joysticks

Rahul Desai, Danyal Sheikh

Abstract: With the help of the multiple arduinos mainly UNO and NANO and multiple HC-05 Bluetooth modules combined with L298N Driver and two DC motor, I will be making a robot car which will be controlled using Joystick which will be connected to arduino NANO.

CS.52 Buck Keeper

Khaja Omer, Harsh Patel, Krzysztof Rusin

Abstract: This project will use multiple Arduino boards connected with a fingerprint sensor to control an external safe. Our design utilizes the fingerprint sensor as a method of input and will trigger a circuit that unlocks a safe for an authorized user.

CS.46 Climate Control System for Plant Growth

Onkar Dangi, Prithvi Ponneboina, Vincent Tamosaitis

Abstract: This project is a climate control system that can support an array of growth boxes, each with unique climate profiles, which can grow plants or mushrooms. Each growth box is controlled by an arduino whose job is to regulate the climate. The user will have access to a control panel with an arduino which is connected to the growth boxes via bluetooth. From the control panel, the user can set the climate profile for each of the connected growth boxes. The current climate conditions will also be sent from the growth boxes to be displayed by the control panel.

CS.32 Color Detector Device

Blanca Puma Bermeo, Daisy Sandoval

Abstract: The idea for this project is to build a device that detects the color of small objects, place objects of the same color in a container and then display the color's name and number of elements. The device will use two arduinos. The first arduino will be in charge of color detecting and placing an object in a container. The second arduino will display the color's name and number of elements. Both arduinos will communicate with each other via serial communication. The input/output devices for this project will be one color sensor, two servo motors, switch, power jack and a LCD display. This project involves original work because this applies the principle of large machines used in the food industry. For instance, machines in the food industry can be used to separate fruit from small debris and leaves.

CS.34 CTA Integrated Bus Tracker

Kenan Arica, Anya Ellis, Darius Grassi

Abstract: The idea of a physical bus tracker is very relevant and needed for city living. Therefore, this integrated bus tracker design through the use of microprocessors could be a simple and more use-effective alternative to what is already used by cities. This project ensures this goal through the use of 3 microprocessors (1 Raspberry Pi 3 and 2 Arduino Unos). The project gives additional functionalities to the user than what is normally seen as it allows for unique user input which pertains to the display and controls.

CS.9 DogGo Automatic Pet Feeder

Arianna Brown, Aisis Julian, Warisa Siewsrichol

Abstract: DoGo is an automatic dog feeder that can be calibrated based on the user's needs. The feeder allows the user to schedule the time/amount for the dog to eat in combination with whether the dog is near the feeder (via motion sensor). When it's nighttime, it automatically turns off to save battery. In addition, the feeder will play a song when it dispenses a treat. With DogGo, owners can create and maintain a feeding schedule with minimal effort. It takes the stress off of scheduling feedings so owners have more time to enjoy their pet.

CS.41 Elemental War (RPS with Earth Fire and Water)

Brian Nguyen, Kent Paglomotan, Zaid Patel, Reggie Sales

Abstract: The days of rock, paper, scissors are boring. We bring an elemental solution to the kids' game to make the simple game a little more tactical. The player chooses their element and based on the conditions that potentially could change mid-match, we determine a winner. The plan is to implement different sensors that can carry a loser into the winner's column through chance and through the temperature, light, and sound all within the room. If not for a draw, the elemental war will be won or lost based on not only the choice but the room's aspects as well.

CS.37 Environmental RC Probe

John Mistica, Nguyen Hoa Pham, Dylan Suga

Abstract: The idea of our project is to develop an environmental probe that surveys the surrounding area for various environmental variables such as gas, temperature, and pressure. The probe will use various sensors to measure these variables and will have wheels that will be powered by a motor. The probe will also scan for obstacles with an ultrasonic sensor and notify the user. The probe will be controlled remotely via PC application using bluetooth. While the probe is being controlled, the user will receive a live video feed so that the probe is able to be navigated remotely.

CS.19 Face tracking and light sensing device

Raimundas Juravicius, Hao Phan, Peter Wu

Abstract: Our project is working on face tracking that will recognize and track faces on the screen. The first Arduino will control the camera and the photoresistor. The second Arduino will control a servo that moves based on outputs from the first Arduino. Third Arduino will be connected to an LCD and 2 LEDs to show detection status + surrounding light level. The LCD will display data readouts for light levels from the photoresistors. If it's too dark the camera wouldn't turn on and waste the camera's power.

CS.28 Garduino

Daniel Aguilar, Mustafa Habeeb

Abstract: The Garduino is a system that enables anyone to succeed in growing and caring for their plant by allowing sensors such as the UV light sensor, soil moisture sensor, SD card reader, temperature/humidity sensor, and a water pump to read in data and take care of your plant. The user will have access to the information taken from the sensors through the SD card by plugging it into a computer and looking at the graphs via Excel. The Arduino will also send current plant information to a phone via Bluetooth when the user requests for it.

CS.3 Get Up and Move: Smart Study Chair

Noemi Andras, Ivana Pavlovic, Ameesha Saxena

Abstract: Do you tend to sit in front of the computer for too long? Do you feel back pain? Lack of movement? If you answered yes to any of these questions, consider "Get Up and Move": a chair that keeps track of your sitting time and lets you know when to take a break. There are photoresistors and load sensors controlled by arduinos attached to the chair to know when you are sitting, as well as a clock that has an adjustable timer. All you need to do is to trust the chair, and let it take care of you.

CS.21 Infinity Cube

Jason Batson, Jazbel Lopez

Abstract: The Infinity Cube is intended to entertain small audiences, as well as provide ambience to any room by reacting to live audio. It is built using acrylic and leverages the properties of one-way mirrors to create the illusion of depth. Using two Arduino's, we decouple the audio input from the cube itself, allowing the cube to be larger since it does not need to be moved, while also providing 2x the processing power and 2x the memory.

CS.22 Intruder Alert!: Simple Arduino Security

Joseph Campanotto, Pawel Rozanski

Abstract: Intruder Alert! is a home security system built using two or more Arduinos. Users take a security element, an Arduino with security hardware such as motion sensors, and then connects it via Bluetooth to a controller Arduino. The controller Arduino acts as the point of interaction for the user, allowing him or her to activate or deactivate security elements. The security elements communicate environmental data to the controller Arduino, which then sends text messages (via Twilio SMS and ethernet) to the user's cell phone based on that data (intruder alert!).

CS.10 Leave That To Me

Siddarth Menon, Aryan Thaman

Abstract: While the smart home has simplified life for many of us, there are still ways to go when it comes to safety and convenience. Fire hazards caused by electronics left turned on for too long are a very real threat, and the amount of energy wasted by leaving lights and fans in use for hours at a stretch is significant. Our device aims to make this process simple and reliable - just tap your phone on that NFC tag at the door, and our device does the rest!

CS.39 LED Battleship

Mohammed Khan, Qasim Mir, Shalin Patel

Abstract: This project aims to build a unique two player interactive battleship game. It would be different from a physical/mobile battleship game as we will be using LED's and other Arduino components. We are going to have the chance to plan, execute, and create something we have grown up playing against each other in a unique way we have never encountered before. We will also learn how to use networking or connecting two clients through a server using the Arduino toolkits that will help us better understand the complexities that Arduino can accomplish.

CS.17 Marco Polo Battle Bots

Jaoudat Karime, Fabian Miranda Corpuz, Biruk Yigzaw

Abstract: Two robotic cars that will use soundwaves and bluetooth technology to play the game Marco Polo where one will lead and the other will follow. Crossing over many dimensions of complexity in engineering.

CS.55 Maze Runner

Mishael Jacob, Ayush Shah

Abstract: We will create a self-driving car that is able to be placed within a maze and find the way out. The car will use ultrasonic sensors to see the path with the most open space and move in that direction until it has successfully solved the maze. The car will also have a visualizer for the sensor system to show how the car is making its decisions.

CS.54 Melodic Musical Motors(MMM)

Husham Alammari, Alex Choi, Andrew Macatangay, Luis Villalobos

Abstract: Melodic Musical Motors is a project where stepper motors are used to produce musical melodies. Four stepper motors will be used to encompass different sound ranges (bass, alto, tenor, soprano). Each stepper motor will be connected to each other via bluetooth. The motors will be able to function asynchronously or synchronously.

CS.27 Mini Pong Arcade Game

Marcos Gonzalez, Jasmine Gutierrez, Haley Orshonsky

Abstract: Our project is a recreation of the arcade game, Pong, using Arduinos. It's the classic game of Pong with our own unique twists on the design. Players will be able to customize their game play experience by changing the colors, sounds, and score of the game.

CS.30 Morse Code Interpreter

Shivam Mehrotra, Jeremy Wong

Abstract: Our project will take input from a user as morse code, then output said morse code as plain text to the other arduino. In order to input the morse code the user will have one button to either press or hold depending on the pattern attempting to be inputted, then when they're done inputting, they'll press a secondary button which will send the data to the other arduino. Our project will only support one arduino being able to input morse code at a time.

CS.16 Multi-Octave Piano Keyboard

Andy Nam, Eugenio Perez, Eivydas Raulynaitis

Abstract: The multi-octave piano keyboard is an Arduino controlled project that unionizes three separate Arduinos and multiple components in order to replicate the functionalities of a traditional piano keyboard. Included with the piano's functionality is 7 standard keys, and 2 octave switches.

CS.2 Music Effect Box

Nam Khanh Le

Abstract: The aim of this project is to create a quick and easy way to modulate sounds or music on the go, and on demand. There are two Arduino UNOs communicating with each other using I2C (Inter-Integrated Circuit), one will be for handling all of the analog inputs, with an LED for indication; and another UNO for handling buttons. These will become signals to either trigger or control chosen effects or parameters in Ableton Live. The buttons would decide whether the audio output contains the effects or not.

CS.50 Nerf Turret

Zain Aamer, Adam Hill, Jonathan Laughton

Abstract: A Nerf turret that is controlled with three Arduinos and two Raspberry Pi's. Input is one Arduino acting as a remote, and a Raspberry Pi connected to a camera, which can automatically track targets. These communicate with an Arduino acting as the receiver, which then translates into directions for a servo controller Arduino which aims and fires the blaster. The second Pi detects a lack of ammo and sends it to the remote to light an LED. The result is a turret that fires based on user input or automatically, using the camera.

CS.12 Noise Pollution Bodyguard

Brian Baez, Hilda Mendoza-Avila

Abstract: Using two Arduinos that communicate with one another, the noise pollution bodyguard is a device that allows a user to track their journey indoors while analyzing sound level of their current location. Sound levels that are considered to be "dangerous" are marked so that the user is aware of them in the future. In addition to this, the user is also given the ability to mark custom locations that they would like to go back to.

CS.44 Personal Radar System

Paul Morgan Iii, Abraham Nakhleh

Abstract: This will be a personal radar that acts as an early warning system to let you know if people are approaching you from behind

CS.48 Playstation with Multiplayer Pac-Man

Viktor Kirillov, Volodymyr Vakhniuk

Abstract: For the project our group has assembled a Playstation, which will launch and run the game the user selects among available games. One of the games will be created by us, which is an extended version of classical Pac-Man. The first extension is the switch from 2D to 3D graphics and the second extension gives two players play together! That means that two random players, who are connected by Wifi, can pass levels together. Other games could be ported to our Playstation easily since the launcher which we created could launch any possible application.

CS.53 Pongino

Jeff Guo, Lorenzo Vicente

Abstract: Pongino is a remake of the old Pong game for Atari but done on Arduinos. Players will be able to move their paddles with physical motions in front of motion sensors in order to hit the ball visualized by LED lights. The game will keep score and announce a winner and a loser at the end of the match.

CS.35 Programmable Cheat Code Lock

Matt Robbins, Kyle Soliva

Abstract: The Programmable Cheat Code Lock's main goal is to create a fun, easy-to-remember, yet complex locking mechanism by using a familiar input system and a simple output screen. Two Arduinos will be used, with one handling input and sending the data to the second Arduino to be read. After calculations are made, output will be displayed letting the user know that their input has been received and read.

CS.1 Remote Controlled Autonomous Obstruction Prevention Robot

Aksa Chacko, Akshaj Uppala

Abstract: In this project, we are creating a Robot on 2 wheels which can move based on the movement of the joystick. The Robot also has the ability to move away from the obstruction in its path; this is done using the ultra-sensors. The ultra-sensors are planned to be rotating 360 degrees in order to sense the direction in which the obstacle is approaching. The systems software immediately overrides the normal functioning of the robot in order to divert the direction of motion of the robot.

CS.11 Rish and Shine-inator

Maher Basrawi, Brad Jimenez, Salvador Jimenez, Christian Romero

Abstract: Our project will be a device that will help the user to wake up in the morning. Our device will have an alarm clock attached to a container. The container will hold a bottle or cup of water. The device will move farther from the user until the alarm is turned off. Once they turn off the alarm, they can take the water from the container to hydrate. The first arduino will deal with the movement of the device. The second arduino will deal with the balance of the water bottle within the container. The third arduino will deal with detecting temperature of the water, and would light up a certain amount of LEDs based on temperature. The last arduino will deal with the alarm clock and the LCD.

CS.13 S.M.A.R.T. RC Car

Damian Charczuk, Raphael Paolo Genova, Wayne Kao, Kai Liang

Abstract: Through this project, we attempt to simulate a version of collision prevention in cars on a smaller scale. To do this, all pre-existing electronics are stripped out of an RC car and replaced with Arduino microcontrollers and sensors. The collision prevention system is realized through the use of ultrasonic distance sensors in the front and the back of the car. Through the use of this system, the car is aware of its surroundings and thus collisions are prevented.

CS.47 Secure Election System

Vivek Ganesan, Kidist Gebremedhin

Abstract: This project involves using 2 Arduinos to build a secure election system. In this project, we will have one Arduino serve as a poll where voters will be able to vote while another Arduino will serve as an administrator for the poll.

CS.29 Security Motion Detection

Abdul Khan, Varang Patel

Abstract: This project will sensor any detection of objects near its range and will send an email or text to the owner. Two arduinos will be used in this project. The sensor will be the input and the notification will be the output. This project can be used to put it in a safe lock in case any one tries to open it or by doors in case any intruder might enter.

CS.26 Sign Language Glove

Romin Pakshir, Mahal Schroeder

Abstract: Our project is called the Sign Language Glove. Using muscle flex sensors fitted onto a glove such that it records the bend angle of each finger, it checks it with the cataloged letters. The signed letter will be displayed on a screen. If the user wishes, they can also push a button and the LCD screen will display a random letter, and the user must match the letter. If the user gets it right, then an LED will light up.

CS.31 Single Player Gear Shifting Racer

Herit Khandwala, Akash Magnadia, Jared Manusig, Matt Ziminski

Abstract: We are planning on creating a Single-player gear shifting car racing game. Four Arduinos linked together will be used to make this game work. Arduino A1 keeps track of the "imaginary" car's rpm and allows the player to shift to the next higher gear. Arduino A2 is used to show the progress of the race in terms of distance traveled by car using LEDs. Arduino A3 is used to start the game, show any needed messages the player should know, and display the car's speed/distance. Finally, Arduino A4 is used for the countdown LED race clock.

CS.40 Smart Home Security System

Syed Hassan, Raymond Li

Abstract: Home invasions are common due to a lack of effective home security systems. Therefore, we intend to create an easy-to-use home security system that alerts the homeowner if their home is accessed without their permission. To do this, the homeowner will activate the security system when they are not expecting anyone to enter the home. If the system is on and someone enters, the buzzer will go off, and the system will transmit information to display the system status on the LCD where the homeowner is in the home.

CS.51 Smart Public Parking Garage

Cristian Gallegos, Amaljosey Johnson, Joshua Thomas

Abstract: Our project will be a smart public parking garage that is highly secure and informative. The garage will use external devices such as an ultrasonic sensor, Bluetooth module, LED screens and a buzzer. The garage will have a security entrance and will have updated information on available parking spots. In the real world, there is too much traffic inside parking garages when people try to search for parking spots when there is none left. The implementation of this in the real world will make people's life easier, secure and faster.

CS.33 Snack Feeder Arm

Dereje Gudeta, Aswini Madugula, Birhanu Mihretie

Abstract: The main objective of this project is to develop an arm-like mini robot that simplifies the task of eating a snack. The robotic arm features a simple sliding design controlled by servo motors, and it is easily controlled with Bluetooth connectivity. The main design aspects of this arm are the distance sensors that calculate the exact movement of the arm that is required to get the robot into the perfect position, as well as the implementation of the pincer to actually transport the snack. The project will feature 3 Arduino interfaces, as well as many input/output devices.

CS.23 Sound Sensitive Ambilight

Mateo Juresic, Valentino Juresic, Justin Trieu

Abstract: The communication of 3 Arduinos to extend the colors on a monitor to the background to give a more immersive experience. It will also take in sound levels to add extra intensity to what the user is playing or watching. The ambilight will include the ability to change modes if the user just wants a specific color or feature as well

CS.49 Temperature Based Power Control

Wyatt Hovde

Abstract: Our project will be an arduino system that will shut down a laptop if it gets too hot. One arduino will be for monitoring the temperature. There will be a bluetooth connection to the laptop and phone that will allow us to monitor the temperature remotely. The second arduino will be used for a motor that will press the power button in order to shut down the computer, also communicated to the phone through bluetooth will be LEDs to monitor the temperature of the unit in question and a button to set the average temperature of the computer for calibration.

CS.20 Temperature Regulator

James Joiner, Nawal Khan

Abstract: Our goal for this project is to create a device that can measure outside temperature, use that temperature to adjust the user's air conditioning/heating device, and alert them to any changes made to their air conditioning/heating temperature. It will use multiple arduino devices that will communicate with each other to accomplish this goal. These devices include: a temperature and humidity module, an LCD display, bluetooth adapters, passive/active buzzers, etc. This device would automate the home temperature process, increasing the ease-of-use for the user.

CS.38 The Drivable Claw

Amjad Ali, Ronnin Padilla

Abstract: This claw will be able to reach anything within its proximity and will do so while moving around on wheels, simultaneously. A controller will control the vehicle wirelessly. It will be possible to "record" previous sequences of claw/vehicle movement so then the vehicle will move autonomously when playing these recordings back. An OLED screen will be the home menu, where you can: set a recording, choose any previous recording to playback, or simply drive the claw vehicle in manual mode.

CS.4 The Self Driving Car with Communicating Screen.

Pushti Dalal, Vedant Majmudar

Abstract: The project is about making a self-driving car based on the ultrasonic radar system. This car will drive in any direction avoiding the obstacles on the way, with communicating what it is doing via a screen. This is just a basic concept of a Self driving car which avoid accidents and lets the owner know what it is doing.



Engineering Expo 2020

CS.24 The Sequencinator 3000

Pat Kolakowski, Krystian Stanuch

Abstract: Our project focuses on a simon says game that requires the player to follow a sequence of LED lights, photoresistor lights, and text displayed on the LCD screen to advance to the next level or if at the last level, win the game.

CS.36 Tic-Tac-Toe

Vittoria Laudando, Spanya Shamasa

Abstract: The purpose of this project is to implement a Tic-tac-Toe game which involves two players competing to get three of their respective symbols in a row to win the game. Both players will take turns selecting their spot on the board and the game will end once Player 1 or Player 2 has gotten three in a row and won the game.

CS.6 Tilt Tilt Revenge!

Luke Austin, Adwait Chavan, Matthew Lange

Abstract: A hardware rhythm-based game utilizing an Arduino Mega and two Arduino Uno's where users will respond to LED light input with a joystick to the beat of your favorite songs and soundtracks! The goal is to move the joystick during a window of time corresponding to the LEDs, and gain as many points as possible. The better you do, the higher your score will increase! Get the highest score, and you can record your name!

CS.7 UICheckin

Tarush Gupta, Akhil Modi

Abstract: Imagine its Final's week and you are already stressed for your exam. You reach the library, run through all the floors just to find a spot to sit and study but you cannot. What if we tell you where you can find a spot? So, here we introduce UICheckin, a feature of the UIC App which tells you where you can find a spot and study peacefully.

CS.25 WakeUp

Lawford Liu, Jason Zhang

Abstract: An alarm device acts as a clock with new functionality implemented by our original works. By setting a "timer" on the device, the alarm would make a loud sound when the time comes. Automatically generate a simple calculation problem from our implemented function. Prompt the user to give the correct answer and stop the alarm. This functionality needs two Arduino to cooperate with a serial connection.



Department of Electrical and Computer Engineering



ECE.7 3D Liquid Dispenser

Radoslaw Jelonkiewicz, Christian Pacheco, Anand Patel, Vivek Patel

Electronics Engineering

Adviser(s): Dr. Renata Revelo Alonso

Abstract: The covid-19 pandemic has shut down institutions of learning throughout the US and around the world, many instructors have been forced to either cancel classroom experiments or figure out how to virtualize the experience. Additionally, when operating physically, minimizing the amount of contact untrained and budding scientists have with dangerous chemicals could improve safety. With this in mind, we have created a remote liquid dispenser which is affordable, easy to assemble and use. Our device can dispense liquid in a 12" x 12" x 12" build area. Our user interface allows the user to control the dispenser remotely in a free roam mode or in a programmatic manner. The latter method can be used to test multiple variations of the same experiment. The dispenser device will have a camera and laser attached to it for users to continuously monitor and observe the experiment.

ECE.5 Air Purifier Bot

Rafeed Hasan, Ishaq Hassen, Keneel Patel, Rohith Pokattel Roy

Consumer Product Development -

Home Automation

Adviser(s): Dr. Ahmet Enis Cetin

Abstract: Particulate matter is a complex mixture of small particles and liquid droplets in the air that can create or aggravate existing health issues such as asthma and or lung function. Particles of less than 5 microns can be caught in the throat and nasal region, while particles measuring 2.5 microns and smaller can be caught in the trachea. Sources of particulate matter can come from construction, smoke, tobacco, cooking odors, dust and pet dander. To combat this issue, we designed an autonomous air purifier robot that can remove contaminants in the air. Furthermore, this product eliminates the need to buy multiple purifiers for a home and will only require very minimal user interaction and maintenance. This product can be marketed as environmentally conscious and medically beneficial.

ECE.14 Assistive Foot Pedal Device

Austin Bunker, Lucas Lawson, Jacob Scott, Teja Thangella

Software Development

Adviser(s): Dr. Miloš Žefran

Abstract: Many computer programs, such as CAD, require the use of a combination of keys to be used effectively. These combinations add mental burdens and inefficiencies that take away from the users' experience. For example, while performing a simple rotation of the object, the user has to simultaneously press multiple keys as well as navigate the mouse in order to rotate the object. For users of this and similar software, our device streamlines the process of digital work. To meet this need, we have created an open source foot-pedal system that allows the user to gain more efficient control of their computer by mapping these key combinations to the movements of the device. Our design is modular and configurable by the user to meet their needs. This was achieved by using low cost and easily sourced components, open source software, and having the entire frame manufactured with 3-D printing.

ECE.10 AutoGreen

Vidhi Jani, Samantha Pacella, Raj Patel, Michael Pekala

Sensor Technology

Adviser(s): Dr. Amit Ranjan Trivedi

Abstract: Community gardens are an integral part of society as they promote sustainability, physical exercise, healthy eating, and food security. Although fostered by entire communities, these gardens are prone to failure due to the inability to maintain a healthy garden and limited knowledge about plant care. We have designed a device to take the guesswork out of plant-based nurturing, so users can deliver optimized care to their gardens. We provide accurate feedback regarding watering, light exposure, temperature, and pH balance of the soil. Users will be able to interact with the device via a phone application and receive live updates on the plant health. In addition, users can review supplementary information about the plants they are growing.

ECE.22 Automated Window System

Russ Boyer, Sashaank Prathap, Signory Somsavath, Bart Wojtowicz

*Consumer Product Development -
Home Automation*

Adviser(s): Dr. James Kosmach

Abstract: Since the turn of the decade, homes have turned towards automation and smarter technologies. Google Nest, being an example, is allowing for home life to become a lot smarter and convenient. From smart thermostats to security cameras, the slow progression towards a fully automated smart home is inevitable, but there's still room for more products to be made. When surveying the market, we realized there aren't many options for automated windows. For this reason, we have developed an automated window system that can open and close windows remotely via a phone application. The system will consist of a temperature, humidity, and rain sensor. With these sensors, the user will be able to customize how their window will open and close automatically. Say if humidity levels become too high, the temperature outside becomes too cold, or it starts raining, the window will close/open as you see fit. The user will have complete control over closing/opening their window, without having to manually do so themselves. This system will bring window automation to a new level.

ECE.18 Autonomous Alexa

Michael Banasik, Karim Eltahawy, Evan Gruzlewski, Kishan Patel

*Consumer Product Development -
Home Automation*

Adviser(s): Dr. Ahmet Enis Cetin

Abstract: Home assistant devices, such as Amazon Echo and Google Home, are stationary devices that are intended to be used in a singular room. Users who wish to expand their voice control within their home are generally left to purchase several devices to distribute throughout their home. Our project was to build a rechargeable, easy-to-use robot with the ability to follow a user, providing power and, most importantly, mobility to users' home assistant devices. This is achieved by having a docking bay atop the robot that provides a stable platform to attach devices, while also acting as a power supply. The device, sporting dual USB ports and an AC outlet, provides broad device compatibility, including phones, speakers, and even laptops!

ECE.2 Autonomous Snow Removal Robot

Mohammed Arshad, Shahzeb Naveed, Mbresa Tali, Marius Zavistanavicius

Robotics Engineering

Adviser(s): Dr. Igor Paprotny

Abstract: The task of removing snow is inconvenient and can be dangerous. That is why we have created the Autonomous Snow Removal Robot (AnSRR). It is designed to displace snow during periods of snowfall to keep the user's driveway clear and ice-less. The user simply needs to set the perimeter of their driveway and the AnSRR will sweep the area before their morning commute. In addition, we programmed a simulation that illustrates how the robot creates a map of the driveway, using LiDAR, during the setup phase and avoids obstacles, using LiDAR and ultrasonic sensors, during the snow clearing phase. IoT connectivity and a friendly user interface will be developed in the future.

ECE.3 Blindspot Detection System for Motorcycles

Watheq Bannosha, Jonathan Hanches, Sahar Muheisen, Vinh Pham

Signal Processing

Adviser(s): Dr. Vahe Caliskan

Abstract: In 2019, over 5,000 fatalities and almost 90,000 injuries occurred due to motorcycle related accidents, and these numbers rise every year. Many of these accidents are caused by motorists not being able to see other passing vehicles. In order to reduce the accident count and aid motorcycle safety, we have developed a portable, modular, and universal blindspot detection system called the "Oculus Veritatis". This device will allow the motorcyclists to become more aware of their surroundings by using sensors to collect data of vehicles passing through their blindspot area and displaying the feedback to the rider with LEDs. We hope that we can reduce the number of fatalities and injuries by providing motorists with a sixth sense.

ECE.24 Communication Aid for the Elderly

Shikan Lian, Zhuoyi Rong, Di Yang, Yiming Zeng

*Consumer Product Development -
Wellness Devices*

Adviser(s): Dr. Vladimir Goncharoff

Abstract: Personal communication technology has undergone an unprecedented boom in recent decades, as people routinely send email, text messages, and post on social media platforms such as Facebook. However, many elderly people who have difficulty in operating a computer or smartphone are left behind. Even the simple task of sending or receiving text messages may be difficult for them. With this in mind, we developed an easy-to-use text-messaging device for the elderly. Whenever text messages are received they are displayed in large font on a display, and the sound of a bell is produced. An elderly person can reply to yes/no questions within the text message by pressing a single button. Various shortcut keys to send standard messages, along with a simple speech-to-text interface, are also provided. Optionally, a necklace with a help button can be used to trigger the sending of an emergency text message to a pre-stored number. Our system is based on a Raspberry Pi microcomputer with a WiFi connection to the Internet, and an online service that provides an interface to text messaging. A Bluetooth connection ties the portable "panic button necklace" to the Raspberry Pi.

ECE.30 Design of an IoT Vital Signs Monitor

Qinzheng Wang, Pengce Wang, Zhiyi Zhang

*Consumer Product Development -
End User Medical Devices*

Adviser(s): Dr. Vladimir Goncharoff

Abstract: Most blood pressure detectors on the market are large, cumbersome devices that are difficult to carry around with you. For this reason, oftentimes when hypertensive individuals feel uncomfortable at work they cannot immediately determine whether this is the result of abnormal blood pressure. Left undetected, such events could signal a life-threatening condition. The goal of our project is to enable the continuous monitoring of a person's vital signs using a wearable monitor. Our device consists of a vital signs measurement unit that a person wears on their wrist. This unit measures blood pressure, temperature, and heart rate. A wireless Bluetooth connection sends the data to a local monitor unit, which shows the vital signs data for the user to see. The information is simultaneously sent over a WiFi connection to the Internet, where the same vital signs are displayed as a web page for authorized users to access. Hence, our system falls in the category of "IoT" (Internet of Things), and makes it possible for doctors to check on their patients without having to make house calls. It also allows concerned family members to see the state of their loved one whom they are not able to visit that day.

ECE.4 EcoSense

Jinbin Cao, Yaner Chen, Khuma Magar, Dustin Rahier

Sensor Technology

Adviser(s): Dr. Zhao Zhang

Abstract: Today, the smart agriculture monitoring market is high in price, high in power consumption and requires a complex operating system. Therefore, our team has designed an optimized system, Ecosense, that is energy-efficient, low-cost, and adaptable. Our system uses a sensor node device that incorporates humidity, UV, and temperature sensors. The sensor nodes use an Ad hoc network called Zigbee to connect and communicate with each other in an adjustable network range, and then send back the collected environmental data through Bluetooth to the user. To further optimize the cost and power, we designed a PCB that incorporates sensors and the power supply port that are crucial for the project. As we continue to develop this project, we hope to innovate agricultural monitoring systems and make them widely available.

ECE.12 Electric Skateboard

Eduardo Espinosa, Victor Ortega, Federico Palacios, Alvin Paras

Electronics Engineering

Adviser(s): Dr. Miloš Žefran

Abstract: New methods of transportation such as electric skateboards and scooters have changed the way people commute. However, once the weather becomes colder, it becomes apparent that electric skateboards and scooters aren't designed to operate in these weather conditions. Icy roads, precipitation, and lower temperatures can make these devices inoperable or even dangerous. We have designed an electric skateboard that can be used during harsh winter conditions and have it work all year round. By developing a skateboard that has a built-in winter mode and enclosures to help operate in colder conditions, people can keep riding their electric skateboard without having to worry about the weather. We have implemented weatherized and protected components, four-wheel drive, and safety features like awareness illumination to create a reliable electric skateboard that users can use.

ECE.13 Emergency Alert System for Everyone (E.A.S.E.)

Hugo Cruz Lopez, Joshua Gligor, Ahmed Kamara, Branimir Maximov

*Consumer Product Development -
Wellness Devices*

Adviser(s): Dr. Ahmet Enis Cetin

Abstract: Over 3,000 people die in a house fire every year. House fires account for 78% of all deaths relating to fire incidents. Moreover, individuals who are deaf/blind are most at risk of harm during the event of a house fire. To help make it safer for them, we created the Emergency Alert System for Everyone (E.A.S.E.). E.A.S.E is a wearable alert system that vibrates to warn the wearer when a fire or gas is detected by our monitoring system. Our wearable device is affordable, reliable, light weight, and easy to use. E.A.S.E. aims to bridge the gap that people with disabilities face in the event of a house fire.

ECE.1 Emperor Mods

Aljo Abraham, Donald Jones, Vaishnavi Medikundam, Francisco Ponce Gallegos

Electronics Engineering

Adviser(s): Dr. James Kosmach

Abstract: Bicycling is a popular mode of transportation and a means to exercise; however, dangers do exist. Per the Chicago Bike Crash Report, there were around 9,000 bicyclist injuries in the span of five years. The report indicates crashes are caused by vehicle collisions and a lack of brake lights on bicycles. Our goal is to keep cyclists safe at an affordable price and to combat the dangers of cycling. We have designed a turn signal indication system, a brake light system, a rear object detection system, and a smartphone application. The turn signal and brake indication system controls are integrated onto the handlebar so that the rider never has to remove their hands. These systems ensure the rider is visible and predictable by motorists. The handlebar control system will use Bluetooth to trigger the turn signals. When a user presses either the left or right turn signal push button it will blink the turn signal LEDs. The brake light LED will turn on when a user touches the brake lever touch sensor. Finally, the rear object detection system detects whether a car is behind the rider and alerts the user if there is one. When these devices are active, except for rear object detection, their use is displayed on the smartphone application. Due to the modularity of this project, the user has the option of which system they would like to purchase in order to stay safe while they ride.

ECE.21 Fern, automated data collection and analysis for greenhouse researchers

Peter Chen, Robert Jaydos, Cheldine Ntsoumou, Daniella Rodriguez, Jessica Vargas

Sensor Technology

Adviser(s): Dr. Ning Jin

Abstract: The aim of our project is to develop an automated data acquisition system for greenhouse researchers. These researchers often monitor hundreds of plants at the same time so collecting data can be tedious. Our product, Fern, alleviates the burden of manual data collection through automation. Fern is an open-sourced platform that is economical and easily adaptable to user preferences. Our platform supports a variety of sensors capable of collecting both environmental and plant data. The collected data is then displayed on a graphical user interface, and through this interface, researchers can interact with the data in two main fashions. First, the user can perform statistical analysis on the collected data using built-in methods, and second, the user can choose how to display the data in visualizations like graphs and charts. Lastly, for cross-platform capabilities, the collected data can be stored in a csv file and then exported to spreadsheet softwares such as Microsoft Excel.

ECE.15 Fire Frequency by The JKObjective

Jonás Chavez, Kathleen Mae Galicia, Olivia Kagan

*Consumer Product Development -
Home Automation*

Adviser(s): Dr. Vahe Caliskan

Abstract: Fire can cause serious damage to a home, but smoke, chemicals, and water have proven to be even more damaging. Furthermore, according to the National Fire Protection Association, residential fires in the United States are the most common type of building fires and are the leading cause of fire-related death and injury as of 2017. Our goals are to help contain small fires before they get out of control and to minimize the amount of water damage from fire fighting. Our team created a device, Fire Frequency, that uses a combination of low-frequency sound waves between 50 and 70 Hz to extinguish a fire. This innovative method of extinguishing fires will help homeowners to have better, safer control over fire accidents. To demonstrate our project, we successfully extinguished a candle from a distance of 1ft. In the future, we plan to make the device accessible for multi-story buildings to extend the area of extinguishing and increase the size of the fire that can be extinguished using our device.

ECE.9 Focal

Adrian Klapisz, Chloetene Malone, Richard Remigoso, Isacc Rojas

*Consumer Product Development -
End User Medical Devices*

Adviser(s): Dr. Hananeh Esmailbeigi

Abstract: Recent studies suggest that there is a strong positive correlation of young adults' mobile phone usage and mental health degradation, according to the National Center for Biotechnology Incorporation. Studies also indicate that modern life constantly bombards late adolescents with news and social media pressure, thus increasing depressive symptoms and loneliness. Considering that 81% of Americans own a smartphone, the rate of smart phone related depression is projected to grow. Additionally, the COVID-19 pandemic has emphasized the role of smartphones in our life. Their association with information-related stress is more vivid. We aim to address this issue by introducing Focal. Focal is a compact phone grip that attaches to any smart phone. This device monitors signs associated with stress and alerts the user when they reach a stress threshold. Focal tracks vital signs such as heart rate, skin temperature, and the tightness of the user's grip. The data is stored by Focal's app and tracks the events of stress over time, both by graphical data and a user-input time-stamped journal. Focal's application is programmed to execute specific commands to output graphical displays of stress levels based on an algorithm. The user can then pinpoint at what time they were most stressed and use the data as training to avoid becoming stressed in the future. Our device also has the capability to track blood oxygenation levels. A potential application for this device would be to help those suffering from COVID-19 or respiratory ailments by allowing them to know when they need to see a medical professional due to respiratory issues. These capabilities combined create a device that we hope will lead to reduction of daily stress with our small device called Focal.

ECE.31 Internet of Things People Counter

Jiaxing Chen, Dongyu Li, Hongbin Sun, Yi Zhang

Signal Processing

Adviser(s): Dr. Vladimir Goncharoff

Abstract: Our team sought to design a video-based system that monitors the number of people in large and medium-sized places to assist managers and security teams in redistributing human resources. The need for such a system exists in most retail stores. For example, queues of people often form at service counters and point-of-sale terminals, waiting to be served. Management would like to respond quickly by sending more employees to service the waiting customers. The system that we designed automatically analyzes a video camera signal, detects and counts the number of people in real-time, and alerts a central computer when a crowd exceeding a specified number has gathered. Our system is designed to have many such units, with each unit doing its own image signal processing so as to be easily expandable. The communications between each camera node and the central monitoring node is according to the Internet-Of-Things model, where each camera node is accessed via WiFi. The most challenging aspect of our project was to detect the number of people seen in an image: due to inherent attributes of biological characteristics, human beings generally look the same but on closer inspection have significant individual differences. We found that detecting faces is the easiest way to identify the presence of people in an image. For this task, we used a Multi-Task Cascaded Convolutional Neural Network (MTCNN) composed of a PNet (Proposal Network), RNet (Refinement Network) and ONet (Output Network) for face detection and human detection. The neural network algorithms will easily run in Python programming language on a dedicated Raspberry Pi computer.

ECE.25 'Knock-Knock' Message Board

Yi Liu, Yuchen Tang, Wuji Zhang, Haiyu Zhang

*Consumer Product Development -
Wellness Devices*

Adviser(s): Dr. Vladimir Goncharoff

Abstract: Our project implements a "Digital Message Board:" its primary function is to receive text messages, and display the text on a monitor. For example, when mounted on the door to an office, the message board may be used to announce that the occupant of that office will be late in arriving due to traffic congestion on the way to work. This is the default mode; the time of day is also displayed. The message board has a built-in camera that enables two additional functions: capturing an image of someone knocking on the door and sending it by multimedia mobile service to a specified phone number, and with facial recognition software identifying specific individuals when they come to the door and showing them user-specific announcements. For example, the person whose office it is can send the system daily schedule reminders that appear when his/her face is recognized. We use a Raspberry Pi 4 computer as the main control unit to realize message transmission, facial recognition, and graphical user interface display. The message board receives text messages through WiFi from a cloud-based service provided by Twilio. When the message board receives a message from a mobile phone containing the desired recipient or the group, it decodes the message, compares it with an SQL user database established in advance, and stores it in the corresponding location. If the facial recognition system matches the camera image face with one the database, any messages corresponding to that user will be displayed.

ECE.6 Land Site Survey Machine

Ray Addi, Syeda Ali, Stephanie Gomez-Luz, Jonas Velasquez

Electronics Engineering

Adviser(s): Dr. Renata Revelo Alonso

Abstract: Land surveying is one of the oldest engineering concepts that has been practiced for centuries. Tools and techniques have evolved in order to help land surveyors record the details and topography of terrain. Extreme weather conditions and complex landscapes are the most challenging problems that engineers face when conducting a site survey. We have designed a system that would allow the surveyor to create a topographic image of land from a secure location. The device will be equipped with multiple sensor modules to measure humidity levels, temperature, distance and height from a reference position. These data are wirelessly transmitted to the engineer's computer, a topographic image is created, and all the data is securely stored on an SD card.

ECE.23 Millionaire-Maker: Stock Price Predictor

Yiheng An, Weijie Guan, Jieyu Zhao, Peng Zou

Machine Learning

Adviser(s): Dr. Vladimir Goncharoff

Abstract: The goal of this senior project was to design software algorithms to help investors make buy/sell/hold decisions for a specific stock according to various factors: the historical performance of this and other related stocks, the stock's price-earnings ratio, and current events that may affect the stock price. In order to guide investors' decisions in real-time, we have designed a system having two algorithms: the first is a stock price predictor applying the Long Short-Term Memory (LSTM) neural network architecture, which is often used in the field of deep learning; it is trained on the past months' prices of a group of stocks of interest to predict future trends. We retrain this predictor every day with new data after the market closes. The second algorithm takes prediction results from the first algorithm, and also input from the user regarding current events and whether he/she considers an event to have positive or negative predicted effect on stock price, to produce buy/sell/hold suggestions. Our project is unique in that two predictors are tied together to give user-specific advice based on both user input and general market trends.

ECE.26 Music Generation Based on Dancer Body Pose Detection

Kenkun Liu, Heng Wang, Xingyuan Zhou

Signal Processing

Adviser(s): Dr. Vladimir Goncharoff

Abstract: Automatic human body pose detection is a video processing algorithm that has recently been developed and made available through open-source distribution. Through pattern recognition and machine learning, a computer can detect and recognize the human body skeleton's position in three-dimensions. Our design product applies this algorithm to recognize and track the body poses of a dancer; then, according to the dancer's movements, the system generates synchronized music in response. Note that this is opposite to the usual concept of "dancing": here, the dancer movements control the music instead of responding to it. Because our system needs to accurately track the dancer in real time, we analyze 30 video frames per second; this processing load is manageable while providing the desired body positional data. We use the open source "Open Pose" algorithm to estimate a dancer's body joint locations (including those of the ankles, knees, hips, etc.) in the two-dimensional image coordinate system. 3-D body joint localization is possible, but was deemed unnecessary for the purpose of extracting the data we needed for music control; this way the algorithm was simplified and made to run faster. We apply statistical methods to process the 2-D body joint data, producing rhythm (downbeat timings) and amplitude-of-movement parameters. These parameters then determine what music piece is played, control its speed and what sound effects are added. We used the SOLA (Synchronized Overlap-Add) algorithm to modify the loudness and rhythm of pre-stored music to synchronize it with the dancer. A variable amount of dancer's control of the music can be specified. Our system gives dancers more freedom of expression, since their movements feed back to affect the music they are dancing to.

ECE.11 QuickPark

Tarek Alzeiback, Mirza Baig, Daniel Wator, Amer Zahra

Signal Processing

Adviser(s): Dr. James Kosmach

Abstract: As people are rushing to get on the train, go to class, and/or get to work on time, they spend valuable time waiting for the payment to be processed for their parking spot or they are issued a citation for failure to pay. This is what students and staff from Washington State University were dealing with when it comes to parking. According to Washington State University Transportation Services, they issued ~20,000 parking warnings and citations in both 2018 and 2019. During this year, however, we are currently experiencing a pandemic and having a contactless system will be risk-free for users to use. That is why we have created a parking system that uses license plate recognition called QuickPark. Our system eliminates the need for physical contact when entering and exiting a parking lot or structure. This system would be able to recognize the user via the license plate of their vehicle and allow them entrance to the parking structure. Additionally, it will bill them for their parking and monitor lot usage. QuickPark will be as fast as possible, and it will eliminate the need for physical contact with parking enters and exit. In the future, the prototype could be developed with other features that can notify the user about their car, such as checking the tire pressure, headlights, etc. Furthermore, this system will be applied for other businesses that'll implement it to their own parking structure, such as restaurants, airports, etc.

ECE.29 Ride-Sharing Driver's Assistant

Chengkang Shen, Peiyan Wang, Kai Yan, Chaoyue Zhao

Signal Processing

Adviser(s): Dr. Vladimir Goncharoff

Abstract: In today's society, automobiles have become an indispensable part of life for both business and pleasure. Because of the many hours worked, fatigue presents a great danger to ride-sharing drivers. Owing to that, our group aims to design a device that provides useful feedback by evaluating driver status and surroundings. It also offers a digital record-keeping to assist the professional driver. In order to carry out these functions, there is a camera mounted on the rear-view mirror that continuously captures images of the driver's face and sends them to a dedicated computer for analysis. We have applied the tools of deep learning, neural networks and feature detection to estimate the driver's current state of alertness. Additionally, we use another camera for road detection. To that end, we combine lane line detection and face detection to get a gradient judgment. When a dangerous condition is detected, the driver will be alerted by lively music and audio announcements with different degrees. Furthermore, our system can also keep a GPS log of driving times and locations for those professional drivers who need this service. Finally, we build a security system that if a stranger starts the car, a text message will be sent to the owner's phone.

ECE.27 Simulated Window Visual Interface

Junling Du, Rui Li, Wenjing Li, Chuanyuan Zhang

Signal Processing

Adviser(s): Dr. Vladimir Goncharoff

Abstract: Our project is to simulate the view through a window between two rooms, even though the two rooms may be across the world from one another. A fish-eye lens detects a 180-degree view in the first room, the video is sent to a second room at a different location, where the signal is processed and displayed. The processing is in response to the viewer's head position in the second room, as it shifts position to "look through the window" of the display monitor. The final effect is that an illusion is created to the viewer: as the viewer's head changes position relative to the display monitor (as detected by another camera), the displayed image changes accordingly to produce a realistic effect. The feature of our system is that users can switch the perspective of observation by simply moving their heads, as if they were looking through a window. Technical challenges to be solved included video transmission between the two environments, and real-time image processing to render a specific perspective view in the second room. Applications of our system include single-camera security monitoring, safety monitoring of elderly relatives who live apart from the family, two-way video conferencing, and interesting art displays of views from famous world landmarks.



Engineering Expo 2020

ECE.8 Smart Garbage Bin

Brian Knight, Vishal Parikh, Matthew Reyes

Machine Learning

Adviser(s): Dr. Matthew Alonso

Abstract: People use many products in their day to day life from food and drinks to their favorite thing on amazon. All these items come in a variety of packaging, which could use plastic, aluminum, paper, cardboard, etc. Most of these items go into the garbage even though some of them are recyclable. For example, every day approximately 8 million pieces of plastic pollutants find their way into our oceans. One reason for this is garbage separation and classification. In the USA, there are at most three bins provided to the user and generally no standard for which items go in what bin. Users are confused and this leads to inaccurate sorting or users defaulting to the landfill. Our team is trying to narrow that window and provide help to the user with the garbage sorting process. Therefore, we have developed an innovative and smart solution called the garbage classification module to help improve the overall recycling percentage and help the user of the bin identify the correct category for the garbage. This will drastically improve the accuracy of recyclable products. This is a better solution for communities and the environment.

ECE.20 Smart Home Cooling System

Stephen Anderson, Jakub Brynda, Edgar Flores, Jeremy Osei

*Consumer Product Development -
Home Automation*

Adviser(s): Dr. Vitali Metlushko

Abstract: Air conditioning systems consume a large amount of electricity and are a significant strain on our energy grid. In the United States, air conditioning can account for up to one third of energy usage during the summer. With this in mind, our team has developed an automated system of window fans that will work alongside an existing air conditioning system in order to save energy and keep the house cool. It monitors the outside environment and brings in cool air when the temperature and humidity levels are appropriate. When in operation, this system suspends the home's central air conditioning to reduce energy consumption. Our system uses RF communication to provide a near wireless install. After hard wiring the thermostat module, the user simply plugs in our outdoor monitoring system and attaches our smart outlet to each window fan.

ECE.17 SPD

Joseph Angelo Bote, Jacob Colgan, Przemyslaw Mosior, Ali Munasser

Clinical Medical Devices

Adviser(s): Dr. Vahe Caliskan

Abstract: The opioid crisis is among one of the most serious issues that the US faces. Over 130 people die everyday from opioid overdose. In order to help combat the crisis and provide further safety to patients, we have developed a smart pill dispenser that will prevent patients from accidentally, or intentionally overdosing on opioids. This device allows pharmacies to monitor opioid prescriptions after it has left the pharmacy, dispense the correct amount of medication, alert the user when it is time to take their medication, and provide emergency options if the device malfunctions. This technical method of combating the opioid crisis will help reduce the number of victims due to opioid overdose and abuse.

ECE.16 Start Blocks

Marcos Araque, Sheng Chen, Brian Hernandez, Jonathan Wacker

Software Development

Adviser(s): Dr. Wenjing Rao

Abstract: The demands of private and public industries for new computer science workers have overcome the ability of the American education system to produce them. According to a nationwide study, 60% of teachers say that they do not feel prepared to teach computer science. Additionally, 40-60% of students that enroll in a stem program in college either switch majors or drop out. As a result, we have built a product intended for students of ages 6-12 specifically to teach programming fundamentals. This product teaches students using a "block based programming" approach, in which code is grouped into "blocks" based on their functionality. These "blocks" are the input to control a video game we created. The game is a series of puzzles in which the user must use "code blocks" to maneuver a character to an endpoint while avoiding obstacles. The block based programming approach is an effective tool to teach programming concepts to beginners, and the addition of physical blocks in our product can give a better visual understanding. This will help students learn computing concepts and general programming skills through an exciting, engaging manner and inspire them to pursue work in a related industry.

ECE.19 Virtual Reality Safety Drill

David Cadieux, Connor Greiman, Pratik Jasani, Alex Smith

Software Development

Adviser(s): Dr. James Kosmach

Abstract: Uncontrollable disasters can strike without warning and potentially change everything about our lives in an instant. Schools perform safety drills (fire, earthquake, tornado, etc.) in an effort to prepare posterity for these emergencies, but for various reasons fail to effectively engage the students in the valuable lessons being taught during these drills. Our goals are to simplify, optimize, and standardize the way safety drills are performed and to provide a tool to conduct an interactive, guided safety drill. We have created a virtual reality simulation that guides the user through the process of performing a fire drill in order to familiarize them with the important steps that should be taken in real circumstances. We built this simulation to be performed on any Android smartphone with the use of a basic VR headset, such as Google Cardboard. With this product, we hope to provide an educational tool that better prepares individuals on how to respond in disastrous situations.

ECE.28 ZYNQ System-On-Chip Standalone Bitcoin Miner

Hongjiang Men, Zijing Xiao, Tianqi Yang, Tianheng Zhang

Software Development

Adviser(s): Dr. Vladimir Goncharoff

Abstract: Bitcoin mining is the process of adding transaction records to Bitcoin's public ledger (blockchain) of past transactions, so that currency already spent cannot be re-spent elsewhere. Mining gets its name from the fact that those who are first to solve a computationally-intensive algorithm (known as SHA256), which adds a legitimate transaction record to the blockchain, are rewarded with newly-generated Bitcoins. Both processing speed and energy efficiency are needed to make any money from mining Bitcoin. The goal of this project was to build a standalone device whose sole task is to mine Bitcoin. This device only needs an Ethernet cable and power supply to operate, and uses the Xilinx ZYNQ System-on-Chip (SoC) as its processing unit. The ZYNQ SoC contains both a fully-functional ARM microprocessor and a highly configurable FPGA (Field-Programmable Gate Array), making it suitable for our purpose. The ARM core handles communications with a Bitcoin mining pool, and the FPGA acts as a fast hardware-based processor to execute the SHA256 algorithm. There are already many FPGA-based Bitcoin mining device designs on the market; what makes our design different is that ours does not require a host PC. An alternative reason for working on this project was to learn more about the Xilinx ZYNQ SoC, which may be useful for future employment. We conclude our work by estimating the cost-effectiveness of our system based on the cost of energy consumed vs. Bitcoins earned, and compare it to some other systems that are available in the marketplace.



Department of Mechanical and Industrial Engineering



MIE.7 3PL Facilities Planning

George Cruz, Kasandra Dominguez, Cesar Garcia

Industrial Engineering

Adviser(s): Dr. Michael Brown; Dr. Quintin Williams

Sponsor: BlueGrace Logistics

Abstract: Within the process of providing 3PL services, there is extreme variability in customer and carrier requirements that create delays between process checkpoints and the collection of real time data. A study was conducted with BlueGrace Logistics, an 11-year full service 3PL company, at the Chicago office location. The company has experienced a plateau in sales as well as a struggling employee retention rate. In order to remain competitive and effective in the market, the company must analyze their process from a continuous improvement aspect. With literature review, interviews with various employees, and collected data the construction of an effective floor plan design was generated. Some of the focuses were Department Layout, Workstation Layout, Information Handling Systems, Information Waste Reduction, and Information Flow Improvements. Our research has identified bottlenecks within management and technologies currently in use. This sparked specific interest with carrier sales reps and rate approval. The team is developing a tool to expedite carrier sales rep managers rate approval to collect data on queue time of this task and continued understanding of the 3PL process. A survey will be administered to qualitatively assess the level of importance on comfort and technology for work productivity in a 3PL office workplace. Ergonomics is extremely important in an office setting. Focus was put on human safety and well-being, office environment, and task/tools. A simulation of the company's current process flow was created with Simio software and data collected from all departments. A 3D model is constructed on Simio to visualize current and potential processes. The goal is to create a simulation that matures with the data analysis determined, developed tools, and alternative methods. The study resulted in the development of a system that requires metrics, standards, and automated data to measure and strengthen efficiency within a 3PL office workplace.

MIE.3 Ankle Exoskeleton

Luke Hutchinson, Michael Maye, Peter Moon, Luis Rodriguez

Clinical Medical Devices

Adviser(s): Dr. Myunghee Kim

Sponsor: UIC Rehabilitation Robotics Lab

Abstract: The objective of this design project was to design an improved ankle exoskeleton relative to the currently utilized Caplex EXO-001 product in the UIC Rehabilitation Robotics Laboratory. Humotech® uses a variety of products to assist patients with ankle, hip, and prosthetic rehabilitation. Maximizing the variety of assistance that these exoskeletons can achieve is the goal of the project. Tasked with the improvement of the exoskeleton, a prototype for a new exoskeleton product was designed that moves in 2 degrees of freedom, is adjustable to men's shoe sizes 6-12, weighs 50% less than the Caplex EXO-001, and operates using two Bowden cables from Humotech® actuators. The design was made with an encoder to measure ankle displacement placed on the hinge joint on each side of the exoskeleton. To mitigate the issues that have arisen from previous attempts at making ankle exoskeletons, the shoes used in the device must be cut so the exoskeleton could be flush with the sole of the shoe. Rubber stoppers are put beneath the aluminum supports to lessen the metal vibration felt when walking. Due to the need to cut the shoe, the original design criteria asked for in the initial stages of the design process could not be met. The design created is an adjustable ankle exoskeleton, not a universal one. The shoe costs also balloon the prototyping costs to the point that only 1 shoe can be bought and cut within budget. The prototype made through the design process is a budget solution to many of the problems introduced by the currently utilized ankle exoskeleton.

MIE.45 AutoCap

Ameen Abdallah, Denisse Manzanilla Reyna, Nicholas Mikhail, George Rizk, Edgard Salgado

Manufacturing Engineering

Adviser(s): Jonathan Komperda

Sponsor: Flexco

Abstract: The AutoCap system is an automated pin-die swaging system that aims to help the belt manufacturing company Flexco finish pins in a safer, more efficient, and lower-cost method. The objective was to create an engineering system that could reduce the need for a floor employee to operate a die swaging machine. The previous method included a floor worker manually applying steel caps to the ends of nylon coated steel wire pins, then standing in front of a large die assembly, and manually inserting the capped pins into the die. The AutoCap system breaks down the task into three main sub-assemblies: the cap applicator, pin swaging, and feeding/translation systems. The cap applicator system will apply caps onto the pins automatically, the design repurposes unused components from Flexco to save on cost. The pin swaging system will insert the capped pins into the swaging die without the need for an employee. The feeding and translation system will begin the process, and allow the pins to move from station to station. In the current design, the only need for human interaction is to load the mechanism with pins. This will increase safety and reduce the cost as less human interaction is needed. Conducting research and defining all variables led to adequate methods of accomplishing the main objectives. From all the possible methods, one was chosen based on financial analysis, time study, efficiency, and quality. The AutoCap design, utilizing the three sub-assemblies, was approved by Flexco. Before developing a prototype, the process was simulated using computer-aided design (CAD) software i.e., SolidWorks. Using this software allowed for any possible complication to be investigated before the actual fabrication.

MIE.46 Automated Gluing Process for Freedman Seating Co.

Theo Hoberstorfer, Drew Johnson, Tahmid Murshed, Kush Patel, Hiba Yaso

Manufacturing Engineering

Adviser(s): Jonathan Komperda

Sponsor: Freedman Seating Co.

Abstract: With adhesive bonding being a fundamental component in the manufacturing of seats, the glue application process must be both timely and efficient. The gluing process currently performed at Freedman Seating Co. is a labor-intensive process in which a worker manually applies adhesive onto plastic seat inserts using hand-held sprayers. The procedure is then repeated on the cloth, and the two pieces are successfully joined together. This process takes approximately three minutes to fully complete. Although this procedure adequately ensures that the cloth and insert can be joined, there is room for improvement. Automation of the gluing process would help reduce inconsistencies, glue waste, operating cost, and time. Specifically, automation reduces glue waste by increasing consistency of glue application on the inserts and reduces overspray. In addition, automating the adhesive application process eliminates the need for a repetitive manual spraying procedure which can have a negative impact on the employees' health. Thus, a group of prospective engineers from the University of Illinois at Chicago is tasked with designing a prototype which focuses on the automation of the adhesive application process. With an aim of improving manufacturing efficiency, the prototype features a disposable conveyor belt system, a mounted air hand-actuated hydraulic sprayer, ultrasonic sensors, a control system, and an optional fume hood. When a plastic seat insert is placed onto the machine, it is detected by the ultrasonic sensor and transported into the gluing area. The sprayer then applies an even coating of glue onto the insert for a set period of time. After the proper amount of glue is applied, the plastic insert is transported out where a worker is then able to simply pick it up and join it with the cloth. Overall, this process eliminates the need for a worker to manually spray adhesive onto the inserts, which may potentially cause long term health issues, while reducing inconsistency, glue waste, operating cost, and time.

MIE.59 Autonomous Steam-Leak Detecting Robot

Justin Bolda, Zachary Gurley, Binh Nguyen, Tyler Thompson, Max Tucker

Robotics Engineering

Adviser(s): Jonathan Komperda

Sponsor: Jonathan Komperda

Abstract: Service tunnels, such as the steam tunnels beneath University of Illinois at Chicago's (UIC) campus, present significant safety risks for workers that must conduct inspections. Hazards such as steam, gas, and other dangerous leaks can be present. Additionally, communication via cellular or radio signals are often not possible in underground service tunnels. With these possible safety issues, precautions must be taken that make inspections tedious and time consuming. Safety, efficiency, and cost are essential concerns in engineering design. The proposed automated tunnel inspection robot would address these concerns by using various sensors to determine if service is needed and map any issues that are found. Automating tunnel patrols and identifying issues will be safer and quicker than manned inspections. These implications result in lower preventative maintenance costs. The automated tunnel inspection robot provides a safer, more efficient, and more cost-effective solution to maintaining underground service tunnels.

MIE.17 Autonomously Adjusting Monitor Frame

Cesar Mondragon, Mahamedsufiyan Patel, Zachary Pindras, Luther Tenorio

Consumer Product Development -

Adviser(s): Dr. Michael Brown

Wellness Devices

Sponsor: Bubble Health

Abstract: This project involves the design process towards the prototyping of a mechanism that holds and adjusts the height of a monitor to the eye-level of the enduser. The mechanism will facilitate the use of a linear actuator and necessary system controls with a total of 1 foot of vertical height adjustment and is expected to hold the weight of the monitor, a shelving unit, and an integrated medical unit. The mechanism prototype is for a robot assistant designed by Bubble Health that would require a height adjusting monitor to accommodate different users with varying heights and positions when utilizing the robot.

MIE.5 Bipedal Robot

Alexander Bandurin, Felipe Cabrera, John M. Lajka IV

Robotics Engineering

Adviser(s): Dr. Atif Yardimci

Sponsor: Bhounsule, Pranav

Abstract: This project encompasses designing a cheap bipedal robot that makes use of easily replaceable, open-source materials, and zero-body length. It is small enough to fit into an average backpack and does not rely on wide feet for substantial support at its base. This robot will be used as part of a class curriculum that teaches students about legged robots. Through extensive research and testing, this team developed a design that relies on a crutch mechanism to propel itself forward. It incorporates three legs, two on both sides, and one attached directly to the center of the body. Servo motors are utilized to power the swinging movement of the legs as well as to manipulate the leg retraction mechanism in such a way as to provide foot clearance.

MIE.37 Composite Fuselage Model

Arthur Correa, Ninoshka Friedman, Daniela Quiñónez, Theresa Wilkie

Manufacturing Engineering

Adviser(s): Dr. Atif Yardimci

Sponsor: AIAA

Abstract: This project is part of a competition organized by The Society for the Advancement of Material and Process Engineering (SAMPE) to design and build a fuselage made from composite material, in this case fiberglass. Team 37 from the University of Illinois at Chicago (UIC) will design and develop multiple simulations for this competition. The design proposed in this investigation withholds 500 lbs. of force as a point load on top of the fuselage and undergoes a three-point bending test. Furthermore, the fuselage must deform no more than an inch after the test, which is on par with the regulations established by the SAMPE organization. The investigation presents a final fiberglass composite model to draw conclusions on the most reliable design.

MIE.34 Cordless steam ironing system

Javier Contreras, Nina Svirinovska, Coby Zaguri

*Consumer Product Development -
Home Automation*

Adviser(s): Dr. Atif Yardimci

Sponsor: Dr. Saeed Manafzadeh

Abstract: The project consists of an ironing system where steam comes from the board itself rather than an iron. The board has an attached steam generator which will be distributed evenly to the board through a manifold. The system will have a built in charging station for the iron and still be able to fold up for easy storage.

MIE.21 Drone to Lift 5 lbs

Mohammed Almughalles, Brett Driscoll, Mohammed El Kadi, Kenjo Pallarca, Sam Zeng

Aerospace Engineering

Adviser(s): Dr. Michael Brown

Sponsor: Dr. Michael Brown

Abstract: Nowadays, drone technology has been improving rapidly from the benefit of different sectors. Defense, transport, health and other industries already took advantage of this highly safe and low-cost technology. In this project, the goal is to design and fabricate a remotely controlled drone with a base no larger than 12 square inches, capable of lifting a 5-pound weight that can fly 20 ft. Knowing the three flight maneuvers as pitch, roll, and yaw, the fabricated drone must demonstrate at least two of the three maneuvers. The design process started by team members' brainstorming, then organization of the thoughts into the technical diagram, as learned during Senior Design I lectures. The 5 lbs weight will be lifted by the fabricated drone which requires the careful design and balance between the drone frame blueprint and motor selection. Many frame designs were created in SolidWorks and tested via ANSYS in order to find the design that could best support the desired weight. The frame is fabricated out of an acrylic sheet using a laser cutter. A flight controller (FC) and an electronic speed controller (ESC) will be used to control the different drone flight maneuvers and receive commands via a transmitter and receiver. This project is sponsored by UIC Senior Design --- University of Illinois at Chicago. The design process was done with respect to all criteria defined for this project, with a budget goal of \$250.

MIE.31 Ergonomic Chair

Niha Ahmad, Aparna Chaganti, Daniel Crudele, Sindhuri Dereddy, Tiffany Pickett

Clinical Medical Devices

Adviser(s): Dr. Michael Brown; Dr. Quintin Williams

Sponsor: UI Health Craniofacial Center

Abstract: Our Sponsor is the Craniofacial Center at UI Health and requires a chair that can stabilize any patient needing pre/post-operation pictures for craniofacial reconstruction surgery. The chair must be able to rotate, recline and extend, while holding patients of every age and size. A possible design has a curved back able to move vertically depending on the patient's height, and would stabilize the patient's shoulders. A second design considered was a chair with a back in the shape of an egg with armrests. A third design considered encompassed the idea of a seat that "hugs" the patient, and no armrest but instead two panels that hold the patient inside the chair. The design selected included all three of the above ideas. A bill of materials has been created with the materials needed and labor cost. The team will contact companies that have large 3D printing capabilities to fabricate a model of the chair design.

MIE.27 Formula SAE Front Wing

Faroq Basharahil, Philip Bobrowski, Jeremy Murray, Rene Paz, Joe Stanek

Automotive Engineering

Adviser(s): Dr. Atif Yardimci

Sponsor: Timothy Kuschel, UIC Motorsports

Abstract: The objective of this project is to design a front wing for the UIC Formula vehicle that will create downforce, minimize drag, and balance the moment generated by the rear wing of the vehicle. The front wing must comply with Formula SAE (FSAE) 2020 section T.7 Rules and Regulations. Once mounted the front wing must not deform more than half of the vehicle ride height under max load. Through research and use of the UIUC airfoil database, the team arrived at a decision to use an S-1223 airfoil profile to base wings designs off of. Each member was assigned to construct a different wing design. These wing designs were then simulated in ANSYS and SimScale to compare the various results and decide on a single preliminary design based on the set of metrics established by the team. A wing design was then selected and was simulated in 2D on ANSYS to determine the angles of attack that will optimize performance. The wing is to be manufactured in 2 inch foam wing profiles that will be glued together, that will serve as the core for the carbon fiber wing.

MIE.22 Green Shipping Container

Kelsey Grossart, Taylor Huynh, Mark Prado, Nicole Xiong

Environmental Engineering

Adviser(s): Dr. Michael Brown; Dr. Quintin Williams

Sponsor: Qmulus

Abstract: The goal of this project was to work with Qmulus, which is a “green” startup company, to design its shipping materials that could withstand the normal forces that a container undergoes while in transit for shipping. The team wanted to maximize ruggedness, recyclability, and reusability while prioritizing the elimination of single use plastics and styrofoam. The approach to the problem was devising a means to properly protect the electronic device that was being shipped while maintaining recyclability. The main problem the container would face was the forces that could destroy internal components, misshapen, and/or break external components. The team applied their collective research and knowledge of shipping practices in order to devise a container which could satisfy all the desired requirements. Some of the potential situations considered while designing the container was to withstand being tossed into a vehicle, or dropped onto the ground upon reaching the receivers’ home or business. Many concepts were formulated, but ultimately one design including only cardboard and 100% recyclable tape prevailed above the rest. Overall, this gives hope of a future with 100% recycled shipping and elimination of single use plastics.

MIE.9 High-Speed Conveyor Spray Transfer Efficiency

Roman Flores, Sammer Saadeh, Oleksandr Wojtowicz, Sam Wywrot

Product Improvement

Adviser(s): Dr. Michael Brown

Sponsor: Kyle Bade, Spraying Systems Co.

Abstract: The overall direction of this project was to improve the transfer efficiency of sprayed material onto a high-speed conveyor belt. A high-speed conveyor moving at speeds over 2000 feet per minute creates a thick boundary layer of air above the conveyor surface which can make it difficult for the sprayed material to reach the conveyor surface. This boundary layer can reduce the overall transfer efficiency of a sprayed material to its desired destination. Within industries that utilize large amounts of spray material, improving transfer efficiency can yield massive savings. With different methods already utilized in the industry (without ample evidence supporting their effectiveness), it becomes necessary to quantify the effectiveness as well as design improved methods to increase the transfer efficiency. With this idea in mind, the next problem to address is how to optimally measure these transfer efficiencies, which is also addressed by utilizing and modifying different setups for spray collection off the belt.

MIE.12 Hip Exoskeleton

Mohamad Al-Rekebat, Yusef Issa, Maciej Machaj, Kwok Keung Man, Madeleine Yuh

Clinical Medical Devices

Adviser(s): Dr. Myunghee Kim

Sponsor: University of Illinois at Chicago

Abstract: The objective of this project is to design and fabricate a soft, wearable hip exoskeleton that improves mobility. This technology would help millions of people in the United States suffering from impaired mobility due to stroke, injury, and disability. A hip exoskeleton could also improve strength and reduce fatigue for physical laborers such as soldiers, firefighters, and factory workers. We will use the humotech Caplex Actuator Unit to power cables in the exoskeleton, and the control will be programmed in Matlab Simulink. The Actuator Unit consists of a servo motor, flexible Bowden cable, and control hardware. This setup will enable us to easily optimize control parameters. Additionally, having an off board power source will allow us to produce prototypes and conduct tests much more quickly and cheaply than if we had to design a portable power supply for every design iteration.

MIE.36 IEEE Robot Platform for Midwestern Robotic Design Competition

Micaela Eaton, Xavier Guzman, Matthew Lassak, Emmanuel Munoz

Robotics Engineering

Adviser(s): Dr. Atif Yardimci

Sponsor: IEEE

Abstract: The purpose of this project is to design and build a robust and modular robotic platform for the UIC student chapter of the Institute for Electrical and Electronics Engineers (IEEE) which will compete in the Midwest Robotics Design Competition (MRDC). This platform will be a three wheel system consisting of dual differential drive wheels and a free moving omni-wheel at the rear allowing for enhanced mobility. The platform will have a top cover designed to accept several different mechanisms with minor modifications. Considerations, influenced by competition rules and requirements from IEEE, were taken into the design and overall shape and function of the robot, especially in the conceptualization of the three mechanisms which will be used. Each mechanism is designed to manipulate a specific type of plastic ball (roughly 6 cm in diameter and 5 grams) with speed, simplicity, cost, and ease of fabrication as the main drivers of the design.

MIE.29 Improved Performance of the SAE Formula Vehicle through Oil System Conversion to Dry Sump

Michael Cuzco, Timothy Kuschel, Kyle Larsen, Koji Okutomi, Michael Spiewak

Automotive Engineering

Adviser(s): Jonathan Komperda

Sponsor: Timothy Kuschel, UIC Motorsports

Abstract: The standard oil management system used in most cars is the wet sump oil system. Wet sump oil management performs sufficiently for standard vehicles but is detrimental in high performance vehicles like the Formula vehicle, where issues such as sloshing, oil starvation, and eventual engine failure are possible at high speeds. An alternative oil system is the dry sump oil system, where oil is stored in an external reservoir and is distributed throughout the system as needed. The Society of Automotive Engineers at the University of Illinois at Chicago is implementing the dry sump oil system for the Formula vehicle by creating a test bench apparatus. The dry sump oil system incorporates a modified fuel tank and an oil pan as two main components, with other components serving as connections for the system. The oil pan is welded from waterjet aluminum sheets and contains thin dimensions with pathing to allow for oil distribution to the pickups. Trials conducted demonstrates improved oil distribution at high speeds and reduced sloshing, which provides sufficient performance for eventual dry sump implementation in the Formula vehicle.

MIE.42 Improved Uptime for Accurate Perforating

Konrad Pietka, Vlad Stratulat, Paul Szeliga, Riley Thomas, Emre Yabukoglu

Manufacturing Engineering

Adviser(s): Jonathan Komperda

Sponsor: Victor Fiszer, Dave Cooper

Abstract: Maintaining a productive and efficient environment is of the top priority for many companies. Due to this universal challenge, a local manufacturing company, Accurate Perforating, seeks to improve its perforating and rolling processes. Specifically, the company wishes to reduce the downtime of their post-perforating sheet leveling process. To solve this problem, Accurate Perforating was provided with a simulation that improves workflow and a sheet counter to track productivity. The process simulation software, FlexSim, was utilized to study the current layout as well as create new solutions that meet the company's goals. Another goal was to improve the consistency of the workplace through more accurate counting of completed sheets. This data is then used in FlexSim to simulate the factory layout and perform any optimizations that were possible. By improving the productivity and effectiveness of Accurate Perforating's manufacturing facility, these proposed solutions will improve customer satisfaction and allow Accurate Perforating to increase production capacity. By completing these two objectives, the company's wishes to reduce the downtime of their post-perforating process will be fulfilled.

MIE.41 Linear Induction Motor System for Chicago Hyperloop at UIC

Adam Awawdeh, Tariq Karroumeh, Mohammed Makki, Khadija Mohammed, Quyen Thai

Transportation Engineering

Adviser(s): Dr. Atif Yardimci

Sponsor: UIC's Hyperloop team

Abstract: The Hyperloop concept was proposed by Elon Musk in 2013 as the fifth mode of transportation aiming to drastically reduce traveling duration for a medium length distance and cost of operation, infrastructure and maintenance. To support the concept development, the UIC Hyperloop team was founded to participate in the 2020 Hyperloop Competition. As a senior design project, our team is participating in the competition by working on analysing, improving and developing a linear induction motor (LIM) that is used for propulsion and suspension in a magnetically levitated pod that can travel in a 10km vacuum tube. This report introduces research and studies done on the LIM to achieve specific requirements, optimize performance and overcome all the introduced problems. Three different configurations of the LIM have been presented as 3D/CAD models and compared with each other using simulation programs to determine which is the best design. The final decision was to combine all the configurations in one design to achieve the optimum performance. Our ultimate goal from this project is to help the UIC Hyperloop team selecting the right design specifications and mechanical configurations of the LIM that is going to be assembled on the competition pod prototype and achieve the highest performance.

MIE.20 Liquid Rocket Engine Combustion Chamber

Tessa Burke, Joshua Gaode, James Huffaker

Aerospace Engineering

Adviser(s): Dr. Atif Yardimci

Sponsor: Dr. Patrick Lynch

Abstract: A liquid propellant rocket is the most powerful method of delivering payloads into outer earth orbit. AIAA's goal for this senior design is to continue development of a liquid propellant rocket designed to reach an altitude of 145,000 feet. RP-1 and hydrogen peroxide fluid properties are determined after injection by the injector plate which are used in ANSYS simulations of gas flow. The chamber length is needed to optimize the burn within the combustor, which was approximated using the throat diameter of the nozzle. CFD simulations are modeled in ANSYS Fluent to analyze the performance of the combustion chamber design. The construction of the engine will be made out of 4130 chromoly steel, used in the aerospace industry for its resistance to corrosion and temperature. Due to the extreme heat of combustion, the engine will need to be insulated. A layer of ethylene propylene diene monomer (EPDM) rubber is used as a sacrificial layer to insulate the combustor during firing. The nozzle will be made out of graphite and is designed to be replaceable after each launch.

MIE.19 Machine Learning Assisted Driving

Nathan Afenzer, Abdullah Alqahtani, Jonathon Faraone, David Keleher, Abraham Weisz

Machine Learning

Adviser(s): Dr. Michael Brown

Sponsor: University of Illinois at Chicago

Abstract: The project objective is to design a machine learning program which can process real time computer vision inputs to assist in the preparation of vehicle functions and vehicle user reaction. The approach includes the application of computer vision software through the use of a dashboard mounted camera, feeding real time information through a micro computer to apply a Machine Learning algorithm to process and communicate appropriate functions through mediums such as LED communications to the vehicle user or a Heads Up Display (HUD). The desired output of the program is to better inform and assist the operator's reaction to upcoming road hazards or immediate changes to the external environment, in addition to preparing the vehicle for sudden input functions such as brake charging.

MIE.53 Machine Learning for Semi-Autonomous Maintenance in a Hazardous Environment

Frantisek Akulich, Christiane Alford, Harsh Gupta, Maz Khoshnood, Muhammad Ishaq Memon

Machine Learning

Adviser(s): Jonathan Komperda

Sponsor: Fermi National Accelerator Laboratory

Abstract: Implementing maintenance automation with machine learning (ML) in a hazardous environment can not only save time and money, but also decrease potential human exposure to life threatening work conditions. A robotic system, semi-autonomously performing maintenance on various highly radioactive parts of a newly constructed accelerator beamline enclosure, is a primary objective of Fermi National Accelerator Laboratory. This will be performed specifically for the Neutrinos at the Main Injector (NuMI) Project. The first step in creating this robotic system consists of the creation of a ML assisted device capable of detecting nuts and bolts on the beamline's front window flange. By using an NVIDIA GPU together with a 3D stereo camera, a deep learning (DL) model based on convolutional neural networks (CNN) is trained on a custom dataset of images representing flanges. These images are captured by means of set camera paths from a rendered SolidWorks model, in order to identify nuts and bolts using object detection techniques. The final product will be made up of a model based on SSD Inception V2, the Intel Realsense D435 depth camera, and a Jetson Nano GPU which will house all the necessary software and python scripts for object detection. From preliminary results for TensorBoard, the model trained using the dataset of SolidWorks images can detect nuts and bolts with an average precision and recall of 74.5% and 61% respectively. In practical terms, this translated into the model being able to detect nuts and bolts on a 3D printed flange in real time within a classification score of 80% to 90%. These results not only prove that a model can be accurately trained using rendered CAD image training data, but it also has generalization ability on real world applications by detecting nuts and bolts in a 3D printed flange recorded by a camera in real time. By including depth analysis to the object detection script, the distance between the target object and the camera lens is obtained which can be used to create a 3D coordinate system. The current work forms a basis for future development of ML and DL techniques for robotics applications.

MIE.33 Matching Candidates to Jobs

Justin Clark, Alvaro De La Fuente, Omar Embaby, Tenev Martin, Jiafeng Su

Machine Learning

Adviser(s): Dr. Michael Brown

Sponsor: Dr. Hadis Anahideh

Abstract: This project aimed to help an IE team to expand their knowledge and experience in data mining. With the support of Dr. Hadis, and Dr. Brown, the team consisting of Industrial Engineering students scraped LinkedIn user's profiles and job postings' features such as requirements location, job title, etc. Using the information scraped, the team clustered user profiles and job postings based on similarity and then matched profiles with job postings looking to find the best match. During this process, the team inspected web platforms such as LinkedIn, adapted an existing LinkedIn scraper using Python, and used data mining techniques such as TFIDF and clustering to reach the goal.

MIE.43 Multi-Specimen Prostate Biopsy Device with Integrated Specimen Storage

Jayvan Gonzalez, Luke Hoppensteadt, Marko Novakovic, Jonathan Ramos, Anya Robertson

Clinical Medical Devices

Adviser(s): Jonathan Komperda

Sponsor: Dr. Michael Young

Abstract: Nearly 1 million prostate biopsies are performed in the United States annually. The current prostate biopsy procedure requires an operator to guide the device to the prostate using an ultrasound probe, penetrate the prostate, cut the sample, and remove the device entirely. An assistant then removes the sample from the device and places it in a storage container. To complete the procedure, this process must be repeated approximately 11 times. We present in this work a new design with the objective of reducing the amount of removals from twelve to one to limit the amount of discomfort felt by the patient as it is largely due to the sensation of stretching caused by reinserting the needle. In order to improve procedure efficiency, we present a design which implements a translating storage tray able to contain all twelve of the samples. To move the samples from the needle to the tray, an innovative mechanism is implemented to guide each sample into the storage tray as the needle is manually retracted. The retraction of the needle is done by disengaging the springs controlling the firing mechanism to allow a low force retraction. As the needle retracts, the storage mechanism moves to the next prelabeled storage slot. When the procedure is complete, the tray is removed, and a lid is placed to seal the samples with a preserving fluid to enable secure transport. The prototype presented intends to be a proof of concept of the mechanical systems, not a functional device that conforms to medical standards.

MIE.57 One-Handed Muscle Roller

Mert Aytunur, Dat Cao, Christopher Joseph, Michael Krupa, Abigail Kunze

Clinical Medical Devices

Adviser(s): Dr. Michael Brown

Sponsor: David Singler

Abstract: A muscle roller was designed to allow athletes to roll out their legs, which will relieve tension and prevent cramps and injuries. While most rollers in the market require the user to sit on the floor or use both hands, this new roller requires the use of only one hand. In addition, its size is small enough to be used in confined spaces such as at a desk or table. This product was also designed to be relatively cheap and easy to manufacture.

MIE.23 Pick-n-Place Robot Arm

Gio Eliashvili, Mario Martinez, Dhruven Patel, Valentin Rufino

Robotics Engineering

Adviser(s): Dr. Michael Brown

Sponsor: Dr. Michael Brown

Abstract: Many distribution companies are able to turn a large profit by increasing productivity with the help of machines. Large scale companies such as Amazon and UPS use robotic arms to aid in the transport of packages or products. Robotic arms are able to lift much more weight and continuously work much longer than any person can. People require rest and are limited by their capabilities. With proper maintenance, machines are not bound by this limitation and are able to function much longer than a person. A robotic arm programmed to lift and sort items can be used to save a company a substantial amount of money. The robotic arm can either aid in lifting heavy weighted products, in the movement of a large amount of product at once, or sorting them for future schedules. The team is tasked with designing and creating a working prototype of a small scale robotic arm with functions similar to an actual robotic arm used in a warehouse. The design uses a barcode reader that will identify barcodes on small wooden blocks. Using the barcodes the robotic arm will sort the blocks on a shelf accordingly. The team uses a Fishbone Diagram, Morphological Chart, and a Selection Matrix. The Fishbone Diagram is used to create a Gantt Chart that is used to track the progress of the project. The Morphological Chart is used to identify the objectives of the design. The team uses a Selection Matrix to select the final design of the robotic arm. The design also consists of a two link system. Where the bottom link is attached to a rotating base and the top link is attached to the end effector. The end effector transports a small wooden block. The final design is a fully automated link system with a clean and simple look that will allow for easy maneuvering and no cable tangling.

MIE.24 Portable Thermoelectric Ice Maker

Robert Filec, Corey Goetz, Enrique Miranda, Taylor Tsinonis, Thomas Xiong

*Consumer Product Development -
Home Automation*

Adviser(s): Dr. Michael Brown

Sponsor: Dr. Michael Brown

Abstract: The purpose of this project is to create a portable ice maker that can be carried comfortably and used by campers/backpackers. Constraints given by the sponsor emphasized the weight of the ice maker to be under 5 lbs. The ice maker should be powered by an alternative source such as solar or a campfire and have a filtered water input. The output should be at least 25 cubes an hour once the system is stabilized. Reverse engineering an existing ice maker was vital to discovering what components took up the most size and space, and which could be removed. The team decided to investigate TEC/TEG (Thermoelectric Cooler/Generator) modules to be the heart of the ice making system. These devices are lighter, cheaper, and remove heat as well as create electricity depending on the application of a temperature difference or a power supply. An assembly composed of a blanket of TECs and a heat sink was fabricated to generate power. Through a temperature difference using a heat source placed under the assembly and a cold source in a tray above it, power can be generated. An assembly with a CPU fan, heat sink, TEC module and aluminum plate was placed inside the ice cube container to aid temperature transfer.

MIE.16 Portable Water Filtration

Tashriq Khandaker, Monak Patel, Andrew Salomon, Eric Taylor

Sustainable Engineering

Adviser(s): Dr. Atif Yardimci

Sponsor: Dr. Michael Brown

Abstract: The purpose of this project is to create a portable water filtration system. This device would be able to take stagnant water from puddle and clean the water into something that is drinkable. A hose would be attached to the filters, and a pump would push the water through the system. This system would be available to be used in third-world countries and would be considered a charitable donation. This is both a design project and an integration project, as several parts are bought already assembled, while others (mostly the housing and connective parts) would be designed from the ground up.

MIE.56 Qmulus EV Charger Hardcover Case

Allan Espinal, Benjamin Moreno, Joel Nolasco, Daniel Torres

Consumer Product Development -

Home Automation

Adviser(s): Dr. Michael Brown

Sponsor: Mathew Raymond

Abstract: Qmulus has designed a smart charger adapter that enables personal electric vehicle charging by converting typical charge stations to smart, connected stations. The adapter will be handled by the user which in turn leads to the possibility of damage due to the impact from a drop. A design for an impact case, which should be recyclable as well as withstand a drop of one meter, has been commissioned. The final design will be made from ABS plastic which is an affordable, recyclable plastic that is also known for durability and impact resistance. The cover will feature three parts, a top and bottom, with a snap-fit locking mechanism, and PE foam to align the inside of the cover.

MIE.6 Quadrupedal Robot

Liridona Ashiku, Maha Mohammad, Nicholas Pippin, Eric Silva

Robotics Engineering

Adviser(s): Dr. Atif Yardimci

Sponsor: Bhounsule, Pranav

Abstract: The following design project aims to develop an inexpensive quadrupedal robot capable of being utilized within academic environments to highlight robotic design and manufacturing. The engineered design model adheres to the Dynamic Robust Actuated Passive Ambulation (DRAPA) challenge rules where an established robot must follow dynamic legged locomotion on a flat surface with steady-state repeatable motion. The simulated design is proficient of being easily reproduced using standard accessible building and electronic material costing no more than \$200. The finalized quadrupedal robotic simulation is projected to translate 50 meters through one-degree of planar motion with a speed range of 0.25 m/s – 0.3 m/s autonomously while balancing on two legs at a time. The calculated overall run time reliability of the robot is projected to be upwards of 3 minutes. Through extensive simulation analysis and modeling, the finalized design simulation attests to all intended design specifications.

MIE.54 Radiation Mapping UAV

Michael Dimeo, Michael Kifarkis, Arshad Narmawala, John Stanton, Sebastian Szulakiewicz, Bruce Treska

Aerospace Engineering

Adviser(s): Jonathan Komperda

Sponsor: Fermi National Accelerator Laboratory

Abstract: The NuMI (Neutrinos at the Main Injector) target system delivers a high-energy flux of muon-neutrinos towards a detector more than 800 km away with the goal of better understanding the origin of the universe and its evolution. The target system becomes radioactive because it is irradiated by a sub-Mega-Watt power proton beam. Radiological hazard is one of critical issues for a radiological worker which limits their activities near the target system; however, they often need to quantify the radiation level of the target system manually. Thus, the objective of this project is to reduce their exposure to radiation while improving the accuracy of the measurements. A quadcopter UAV fitted with a radiation dosimeter is ideal to traverse the irradiated environment as it can be deployed in any space and requires minimal hardware integration to the test room. Due to the lack of indoor GPS signal, this UAV utilizes a series of ultrasonic beacons that release several varied-frequency pulses simultaneously in order to localize the drone. The flight controller returns a 3D coordinate tied to the radiation data to understand precisely where radiation spikes are being emitted from the horn. The flight controller stores log data that can be processed into an easily interpretable map of radiation intensities. The radiation map serves as one of the primary references for planning the transportation of the NuMI horn.

MIE.52 Radioactive Roomba

Erich Christ, Keefe Li, Brian Mlot, Ramsha Shahid, Joanna Tan

Environmental Engineering

Adviser(s): Jonathan Komperda

Sponsor: Fermi National Accelerator Laboratory

Abstract: The goal of this project is the development of a robotic cleaning mechanism (RCM) that collects radioactive dust off the floors of the particle accelerator tunnels located at Fermi National Accelerator Laboratory (Fermilab). When the particle accelerator is active, it produces residual radiation that is absorbed by dust in the tunnels becoming a radioactive hazard. After researching dust collection options such as vacuum, tape, and electrostatic surface, it was determined that microfiber cloth was the ideal choice. The final solution is a cleaning mechanism that continuously feeds in new material of microfiber cloth that is pressed against the ground to wipe and absorb dust particles. The cloth is then rolled up and collected for easy disposal. An initial experiment showed that microfiber cloth has a dust capture rate of 85.6% per 5 grams of dust in 9.64 ft² per 1 ft² of cloth and is considered the optimal performance metric. A second experiment using the prototype RCM showed a dust capture rate of 39.0% per 5 grams of dust in 2.95 ft² per 1 ft² of cloth. Utilizing the second experiment's data and extrapolating the cost of building and operating the RCM to the 94,000 ft² facility at Fermilab, the potential five-year savings plan for Fermilab versus their current outsourcing cost is \$298,365. Future iterations would focus on improving the dust capture rate to match that of the first experiment. Such iterations would include adding a dynamically actuating mechanism to the system that presses the cloth to the ground. The refined system would form to the deviations in the floors thereby allowing for a higher dust collection capacity from the RCM. The future end goal of this project is to create a fleet of autonomous RCMs that would clean the tunnels periodically to reduce buildup and overall operation time.

MIE.10 Rainfall Simulation

Aleksander Kowalkowski, Phil McDonough, Jesus Resendiz, Hoa Tran

Product Improvement

Adviser(s): Dr. Michael Brown

Sponsor: Spraying Systems Co.

Abstract: Many devices are expected to operate under rainfall conditions for extended periods of time without suffering significant loss in functionality. As waiting for such conditions to arise naturally is impractical for the purpose of product testing, this imposes the need for a system which can artificially replicate these environmental effects. This paper provides a fresh examination of the issue, with a particular focus on the production of large droplets. Guided by a review of documented literature on the subject of rain simulation, as well as conversations with engineers from Spraying Systems Co., a selection of current and novel solutions were chosen for further testing. By making use of high-speed cameras and imaging software, the resulting droplet size from each concept was measured and compared based on various industry standard metrics. With an aim of generating droplets on the order of 4000 μ m or above, it is clear that common solutions involving full-cone nozzles (such as those manufactured by Spraying Systems, and used in many rain simulators) are not ideal for the task. These methods involve a stream of fluid which atomizes as it passes through the nozzle; however, it is the determination of this report that larger droplet size is achievable if fluid leaves the nozzle in a solid stream and becomes atomized at some point afterward. This can be accomplished by means of a deflection surface oriented normal to the nozzle, against which the fluid stream can break up. Droplet size is also affected by the position, diameter, and texture of this deflection surface, and the specific effects of changing these parameters are explored. While a standard full-cone nozzle was capable of producing droplets with a diameter of 2285 μ m (Dv0.9), this diameter increases to 4387 μ m when using a stationary deflection surface method with the optimal parameters.

MIE.35 Rapid Configuration 3D Printer

Benjamin Hawkinson, Dan Santoyo, Dominique Slezak, Ryan Tris

Consumer Product Development -

Adviser(s): Dr. Atif Yardimci

Home Automation

Sponsor: Luxx Enterprises

Abstract: Select industries have begun to adopt additive manufacturing as a viable alternative to traditional manufacturing methods. One of the major factors that has prevented the widespread adoption of additive manufacturing has been the cycle time required to produce a given object. In order to reduce cycle time, a novel method of additive manufacturing has been developed which employs a rapidly configurable mold cavity. The mold cavity is formed by an addressable pin array. The array is able to configure into a user defined shape, after which a polymer is dispensed and cured within the mold cavity. The part is ejected, and the mold cavity is ready to reconfigure itself. The examined methodology provided a reduction in cycle time and has opened the doors for alternative materials to be utilized. In order to determine the practicality of this method, a prototype has been assembled and tested.

MIE.39 Redesign of Amtrak's HVAC Duct

Baseer Jagirdar, Mohammad Rahman, Joel Thokkudubiyapu, Bilal Vohra

Transportation Engineering

Adviser(s): Jonathan Komperda

Sponsor: Amtrak

Abstract: The goal of this project is to redesign the duct currently in use by Amtrak for its GE P-42 Locomotive Fleet. The first task is to create a new duct design that is manufacturable in-house at Amtrak, rather than purchasing from a third-party. This design would ultimately replace the current design, while maintaining or improving efficiency. The second task is to improve the installation process of the heating, ventilation, and air conditioning (HVAC) unit, so the duct does not get damaged upon installation. Upon reverse engineering the current duct, a 3D model was created on Solidworks and analyzed on Ansys through various computational fluid dynamics tests. Using the current design as a basis, a new design was created using Solidworks Sheetmetal, which is easier to manufacture through a series of bends. Comparing the two designs, the new design yielded a pressure distribution almost identical to that of the existing duct. Regarding the installation method, the previous method of installation included placing the HVAC unit onto the locomotive using a forklift and a guide, who spectated and gave direction to the forklift operator. This method caused damages to the interior of the locomotive, specifically the duct. The new proposition is to add extra material to the side of the duct, which absorbs the damage from the HVAC unit. This extra material will also serve as a funnel to guide the unit in place.

MIE.38 Retrofit Filament Loader for 3D Printers

Maria Bortolaso, Marco Bucio, Emma Hoyt, Jacob Key, Zachary Sablay

Manufacturing Engineering

Adviser(s): Dr. Atif Yardimci

Sponsor: Mark One SRL

Abstract: Just as 2D printers require cartridge replacement after ink levels have depleted, 3D printers require reel replacement after filament is expended. This limitation provides an opportunity for minimizing time to print, as well as reducing operating costs concerning manual labor for conventional filament reel replacement. The Mark One Filament Loader is a retrofit device that removes interruptions to 3D printing operations by serving as an automated filament loading solution. It consists of an inductive sensor that relays information about the presence of filament and signals several motors to run filament through the device housing according to commands provided by an accompanying Arduino. The device is designed to be scalable for other 3D printer models and to accommodate multiple filament sizes for an individual printer. Due to time and resource constraints, the prototype could not be tested for dual reel system integration. Future work can be conducted to finalize the prototype by integrating simultaneous dual reel loading as originally intended. The main considerations for the prototype focused on hardware integration, process automation, and manufacturability.

MIE.25 RoboBubble Pill Dispenser

Liron Fridman, Branden Key, Alex Medina, Anthony Romero, Jason Schlicher

*Consumer Product Development -
End User Medical Devices*

Adviser(s): Dr. Michael Brown

Sponsor: Bubble Health

Abstract: Bubble Health is designing a robot to help elderly people with their daily needs. The objective is to create a delivery mechanism that will eventually be integrated into the overall robot design. The product is to reliably dispense amazon pill packs to the consumer as needed according to the robot software. This has been done by using an electric motor in combination with gears, drive rollers, and an internal body. The overall design is modeled in solidworks and tested in ANSYS for stress analysis. A part of the prototype has been created, however, a final prototype was not complete due to limited access to resources.

MIE.28 SAE Formula Steering Wheel

Federico Cisneros, Matthew Colagrossi, Angel Fernandez, Matthew Jaworski

Automotive Engineering

Adviser(s): Dr. Atif Yardimci

Sponsor: Timothy Kuschel, UIC Motorsports

Abstract: The purpose of this project is to design a new steering wheel for the F20 formula car that the Society of Automotive Engineers (SAE) at the University of Illinois at Chicago (UIC) will be racing this year. The steering wheel is a major component that aids the driver's ability to control the vehicle. The new steering wheel should improve the driver experience by having an ergonomic design, and an intuitive shifting layout. Through market research and consultation with the chief engineer of SAE the shifting mechanism was narrowed down, the team chose the design with the idea of simplicity to limit component failure rate. The steering wheel includes 3 momentary push buttons, two of which are for gear selection in sequential order, and another for launch control.

MIE.2 Safe-Cut Guillotine Trimmer

Phu Au, Diory Gong, Kevin Mallari, David Vallejo, Andrew Vo

Product Improvement

Adviser(s): Dr. Michael Brown

Sponsor: ACCO Brands

Abstract: This project aims to improve safety features of ACCO Brand's guillotine paper trimmer. The goal is to create a safer design through covering the blade while it is not in use but still retaining the guillotine cutting mechanism. The main criteria for the improved designs are that it can cut at least 10 sheets, has a bed height of 60mm, require a maximum force of 25lbs to operate, clamps paper to avoid movement, have a clean cut, and have no exposed blade.

MIE.1 Semi-Automating the Reloading of a Lamination Machine

Umesh Chaudhari, James Jarr, Thomas Mathews, Yingze Sun

Product Improvement

Adviser(s): Dr. Atif Yardimci

Sponsor: ACCO Brands

Abstract: The goal of this project is to redesign Acco Brand's lamination machine, the Ultima 65 in order to load rolls of lamination safely, while reducing the number of steps to load the lamination film. Due to infrequency in changing the film rolls, mishaps in loading the film can occur. The team members developed a system that will enable the user to load the lamination rolls while decreasing the number of steps thus making it easier and more intuitive to load the lamination rolls. The group optimized the design by targeting the part of the operation that required the most steps and user interaction. Using actuators reduces the number of steps and interaction with the lamination machine while retaining safety. The user can load the film and then close the safety guard of the lamination machine which engages the actuators to lock the lamination rolls into place. In order to meet the requirements of friction to prevent the lamination rolls from slipping, rubber is integrated into the supports that hold up the lamination rolls. The actuators will be powered and controlled using the existing power supply and circuitry of the lamination machine. No internal components are rearranged to accommodate design changes rather the new design is bolted on with slight modification to the screw thread locations.

MIE.40 Strand Burner

Michael Baureis, Kyle Mcdougall, Ghanshyam Patel, Aakarsh Srivastav

Aerospace Engineering

Adviser(s): Dr. Atif Yardimci

Sponsor: Dr. Patrick Lynch

Abstract: This project was assigned to the team by the organization AIAA (The American Institute of Aeronautics and Astronautics), the desired objective is to finish the design and setup of a strand burner. A strand burner is an apparatus that is used to collect the burn rates of solid propellant simulated inside a rocket. The strand burner provided to us was pre-assembled and the team was responsible for writing the code to run and collect data from the apparatus as well as create a ventilation and plumbing system. After these tasks were completed, the team then sketched and implemented an electrical wiring system followed by the ventilation systems and the input power setup. After researching and gaining a deeper understanding of the principals and functionality of a strand burner, the team assessed the crucial steps that needed to be taken in order to fulfill the sponsor's expectations of a fully functioning strand burner. The strand burner was successfully completed and satisfied the sponsor's requirements of being able to ignite a piece of propellant and gather the burn rate and pressure readings in the desired analytical format on LabView. The strand burner is now ready to execute its primary objective; effortlessly and autonomously.

MIE.18 Sun Buckets

Christian Mactal, Mateusz Nawrot, Brandon Villanueva

Sustainable Engineering

Adviser(s): Dr. Atif Yardimci

Sponsor: Dr. Matthew Alonso

Abstract: The main purpose of this work is to provide a cooking alternative without the use of fuel resources while maintaining cultural norms and cooking traditions. Over 3 billion people worldwide cook with wood, charcoal, or animal dung which results in open fires and emissions. As a consequence, prolonged exposure to the emission output has contributed to air pollution and more than 4 million premature deaths. To counter this global issue, the use of solar energy is a solution that can provide a safe cooking resource while still respecting all cultures. Our project sponsor, Matthew Alonso, PhD is one of the founders of the Sun Buckets company which produces portable cooking units that store and use solar thermal energy. The objectives of the Sun Buckets senior design project are: improving the device by enhancing the insulation performance, reducing cost, measuring the temperature and pressure of the inner vessel. Currently, the most recent Sun Bucket model uses Superwool material where the insulation makes the outer shell temperature to be 42°C in 25°C ambient temperature indoors. To improve the existing insulation, the target temperature for the outer shell was to be improved to 35°C or less. CAD files were subjected to ANSYS Workbench simulations to model how each material of interest behaved and reacted when subjected to boundary conditions. After rigorous testing, cork turned out to be the best material to use with results yielding 30°C. To change/improve temperature measurement, many different techniques were taken into consideration. Calculating the pressure was the last objective, which would determine the highest pressure before the vessel inside the Sun Buckets deforms. Boundary conditions were taken into consideration in ANSYS static structural simulations. Both worst case and best case scenarios for convection were done as well as 400°C for the charge of the vessel.

MIE.60 Tunnel Robot

Zeyang Lei, Jinrui Ma, Xinnian Wang, Ruiqing Yang, Sirui Yang, Chunming Yang

Robotics Engineering

Adviser(s): Jonathan Komperda

Abstract: Some underground tunnels are filled with pipes carrying steam and refrigerant. Navigating these tunnels to identify leaks or problem areas is inefficient and poses risks to workers. However, the development of robotics and computer technology allows for robots to do these inspections under extreme conditions, safely and efficiently. The goal of the project is to design an automated robot-type device that would work in the underground tunnel of the University of Illinois at Chicago to check for cracks and leaks in pipes carrying steam and refrigerant. NVIDIA's Jetson Nano was chosen as the processor because of its strong information processing capabilities for real-time information processing and sophisticated Python development environments. An infrared camera, along with an ordinary camera, were used to detect leaks and eliminate the influence of other interference factors. The tunnel robot can follow a line in the tunnel, and synchronously record its position, while storing photos identifying problem areas, such as pipe leaks. Due to the harsh environment of the underground tunnel, the off-road performance, heat resistance, and moisture resistance of the electronic device were also taken into account. Accomplishing these goals will reduce the risks associated to working in underground pipeline tunnels and reduce costs by informing workers of necessary preventative maintenance.

MIE.30 United Airlines Calibration Lab Redesign

Brandon Dang, Bryan Eisenhard, Jason Hermoso, Minji Kim, Bryan Macatangay

Industrial Engineering

Adviser(s): Dr. Michael Brown

Sponsor: United Airlines

Abstract: United Airlines contacted the University of Illinois at Chicago for a student team to assist them in a project to design a floor plan for their Vickery Calibration Lab and its expansion into the departing carpet shop, that maximizes throughput of calibrated tools while maintaining a safe environment for the technicians to work in. The opening of new space will allow the student team to arrange disciplines in multiple layouts and compare them in order to determine the most efficient utilization of space. With the high-level process flow for calibrating tools already established, the students must determine where to arrange the calibration disciplines in order to establish an efficient tool flow throughout the facility. Due to technicians expressing their frustrations with the workspace allotted to them, we are further tasked to apply ergonomic principles to every workstation, suggesting new designs that will accommodate the technicians' needs. In addition, the Vickery lab is looking to measure time of tool usage and idleness more accurately through the use of RFID technology. The implementation of barcode scanners demonstrates to them the indicators of their performance level and then could apply benchmarks to their throughput and other indicators they are looking to measure.



Engineering Expo 2020

MIE.32 YRC Equipment Optimization

Roxanna Ariza, Ricardo Donoso, Julia Leon, Patryk Stalmach, Anthony Vuolo

Industrial Engineering

Adviser(s): Dr. Michael Brown

Sponsor: YRC Freight

Abstract: YRC Freight, also now known as YRC worldwide is a less-than-truckload (LTL) transportation industry. Less-than-truckload is used to transport smaller sections of freight that do not fill an entire truck. Many other transportation companies do not do LTL transportation because it is much more difficult to make a profit that way. YRC breaks up their locations into two different entities. They have distribution centers (DC) and end-of-the-lines (EOL). DCs are much bigger than EOLs and move a higher volume of shipments. DCs are usually located right outside of major cities, while EOLs are located in suburbs and rural areas. Each DC and EOL has a certain amount of equipment. YRC is looking to expand their locations. They want to see if a program can be created to see the maximum amount of equipment at a new DC or EOL. The team, with the data provided, will analyze the amount of equipment and deliveries to optimize the amount of equipment a new DC or EOL.



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